



US012398732B2

(12) **United States Patent**
Zhu

(10) **Patent No.:** **US 12,398,732 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **PORTABLE FAN NECK FAN AND FAN ASSEMBLY**

F04D 25/0673; F04D 25/0606; F04D 29/24; F04D 29/441; F24F 2221/138; A41D 13/0025; A42B 3/286; H02K 5/22

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USPC 417/423.15
See application file for complete search history.

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **19/007,982**

(22) Filed: **Jan. 2, 2025**

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(65) **Prior Publication Data**

US 2025/0237232 A1 Jul. 24, 2025

Related U.S. Application Data

(63) Continuation-in-part of application No. 18/933,625, filed on Oct. 31, 2024, and a continuation-in-part of application No. 18/421,213, filed on Jan. 24, 2024, now abandoned.

Primary Examiner — Kenneth J Hansen

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(51) **Int. Cl.**

F04D 29/52 (2006.01)
F04D 19/00 (2006.01)
F04D 25/06 (2006.01)
F04D 25/16 (2006.01)
F04D 29/64 (2006.01)

(52) **U.S. Cl.**

CPC **F04D 29/522** (2013.01); **F04D 19/002** (2013.01); **F04D 25/0673** (2013.01); **F04D 25/166** (2013.01); **F04D 29/646** (2013.01)

(58) **Field of Classification Search**

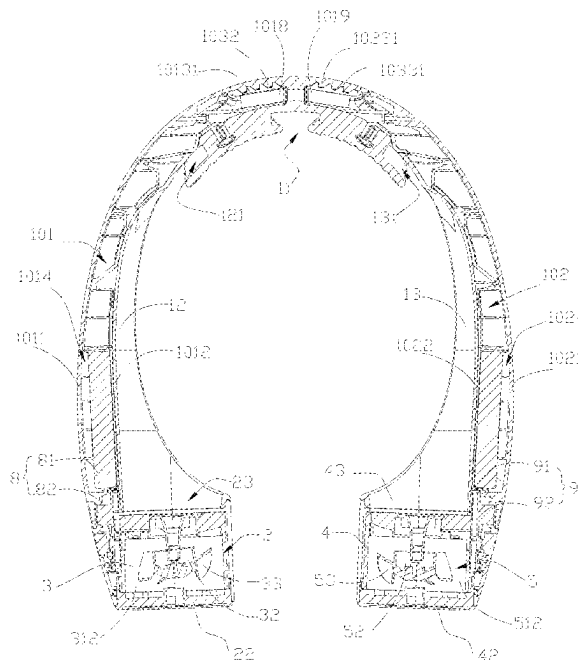
CPC F04D 19/002; F04D 25/084; F04D 25/166;

(57)

ABSTRACT

The present invention provides a portable fan, a neck fan and a fan assembly. A portable fan includes a first main body portion and a first fan assembly. The first main body portion has an arc-shaped structure, and includes a concave first air guide path positioned on a surface of one side of the first main body portion and extended along an extending direction of the arc-shaped structure. The first fan assembly is arranged at one end of the first main body portion and is configured to blow air towards the first air guide path, and the first air guide path is configured to guide air of the first fan assembly to one side of the first main body portion and one end of the first main body portion away from the first fan assembly.

20 Claims, 22 Drawing Sheets



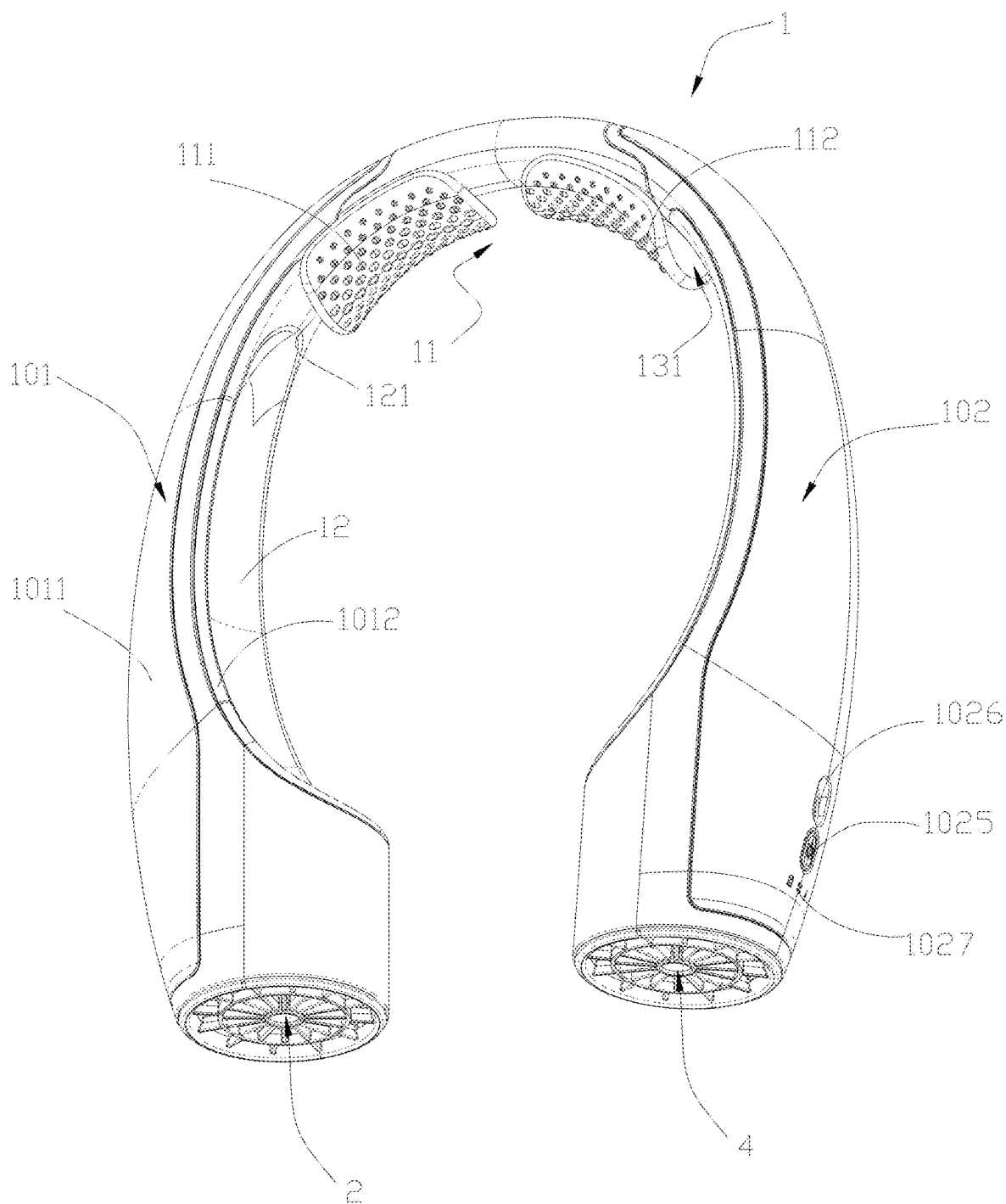


FIG. 1

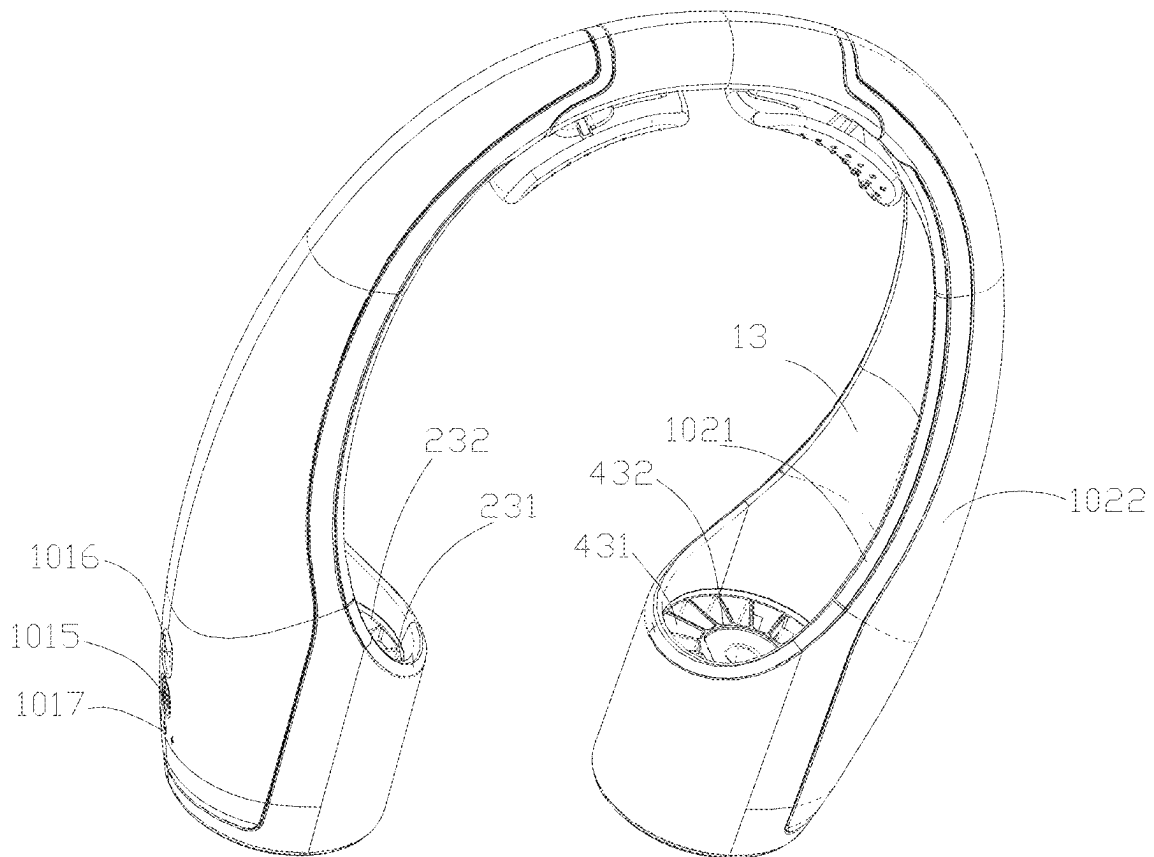


FIG. 2

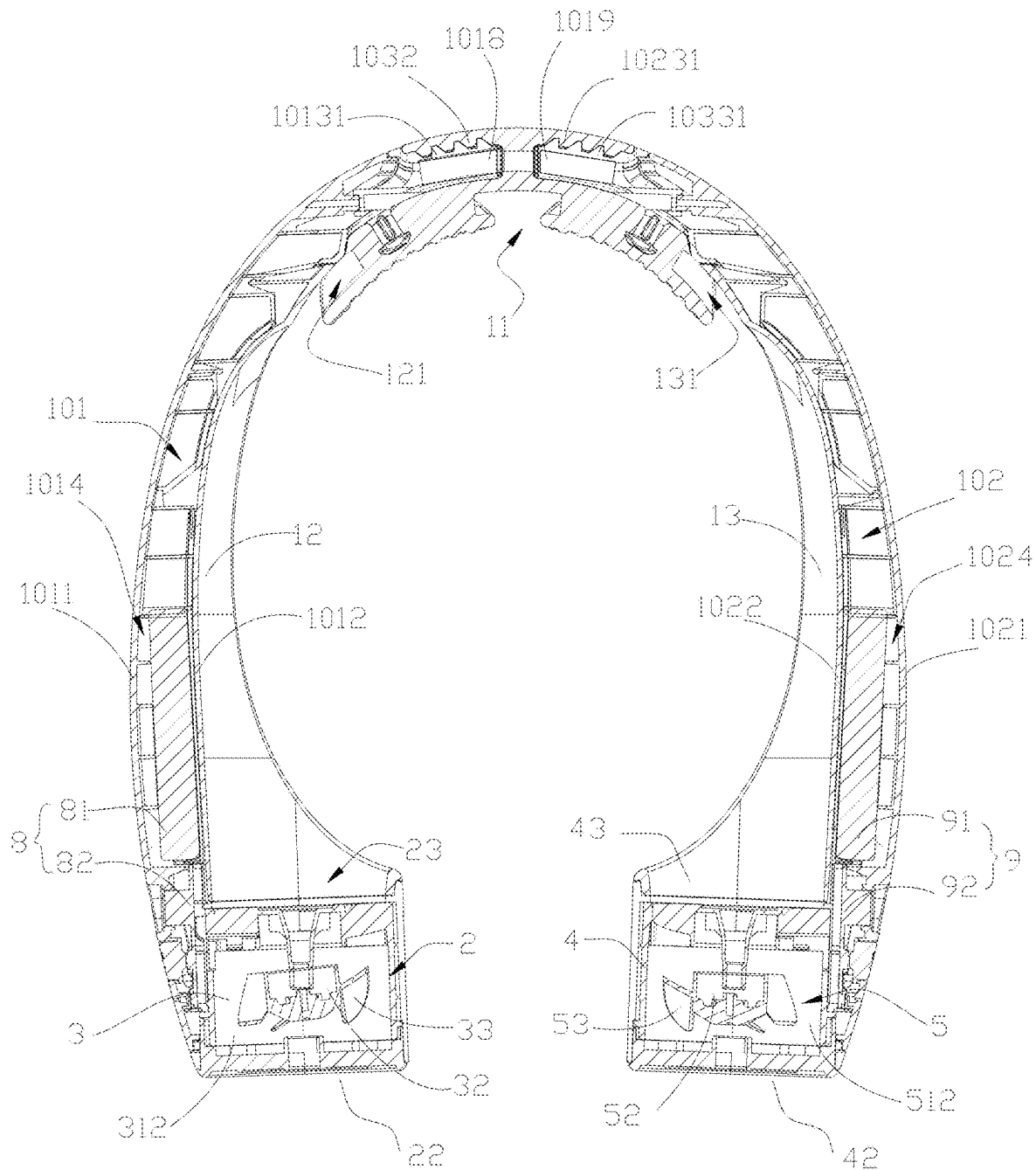


FIG. 3

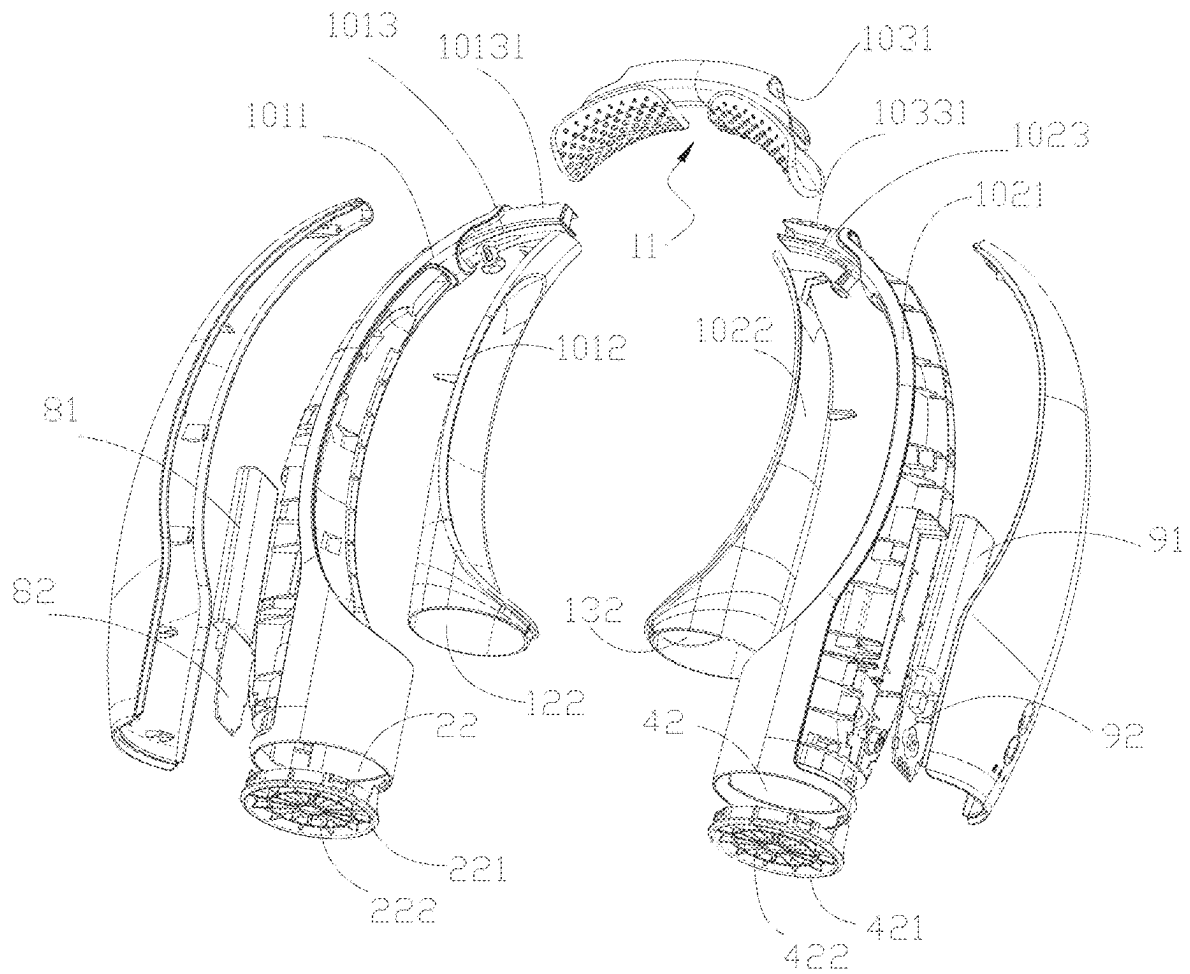


FIG. 4

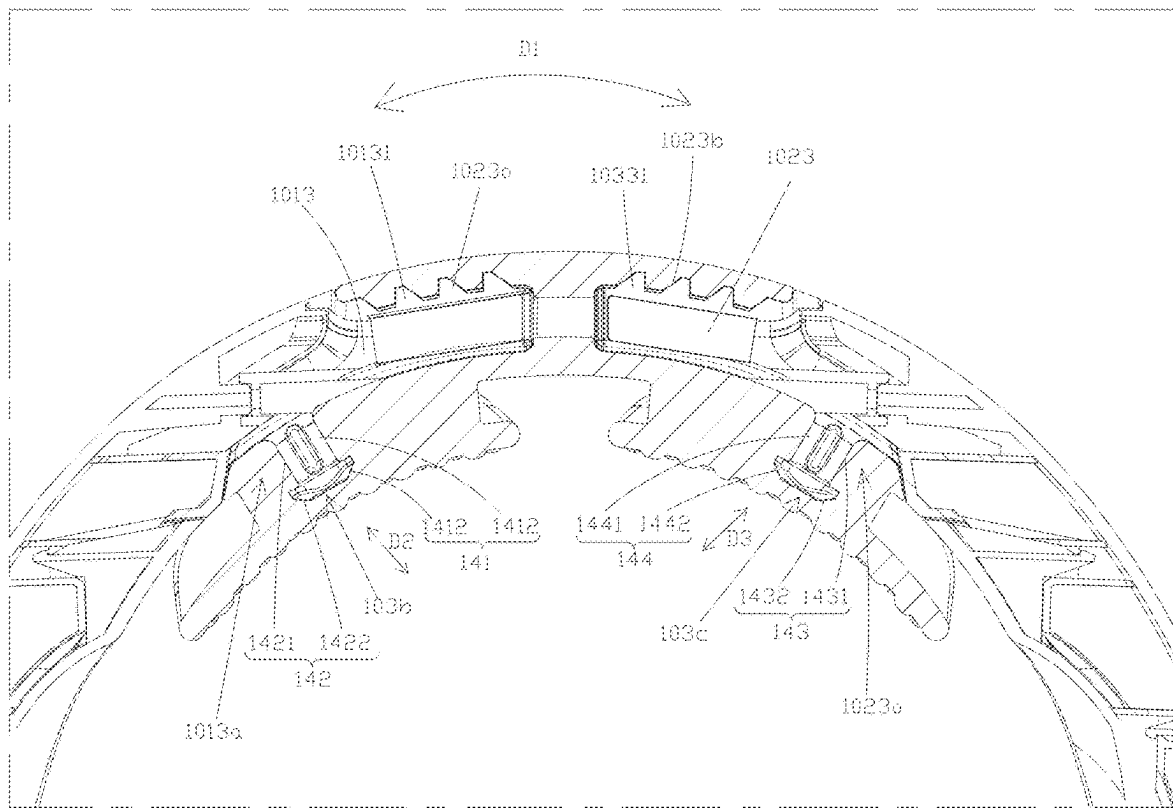


FIG. 5

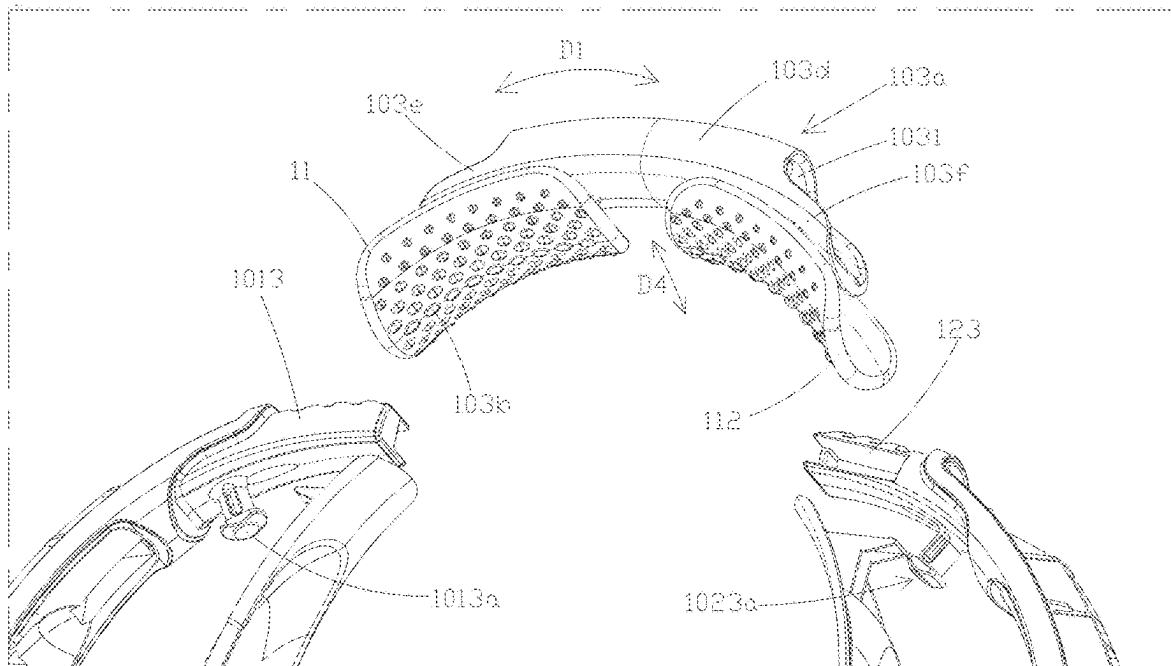


FIG. 6

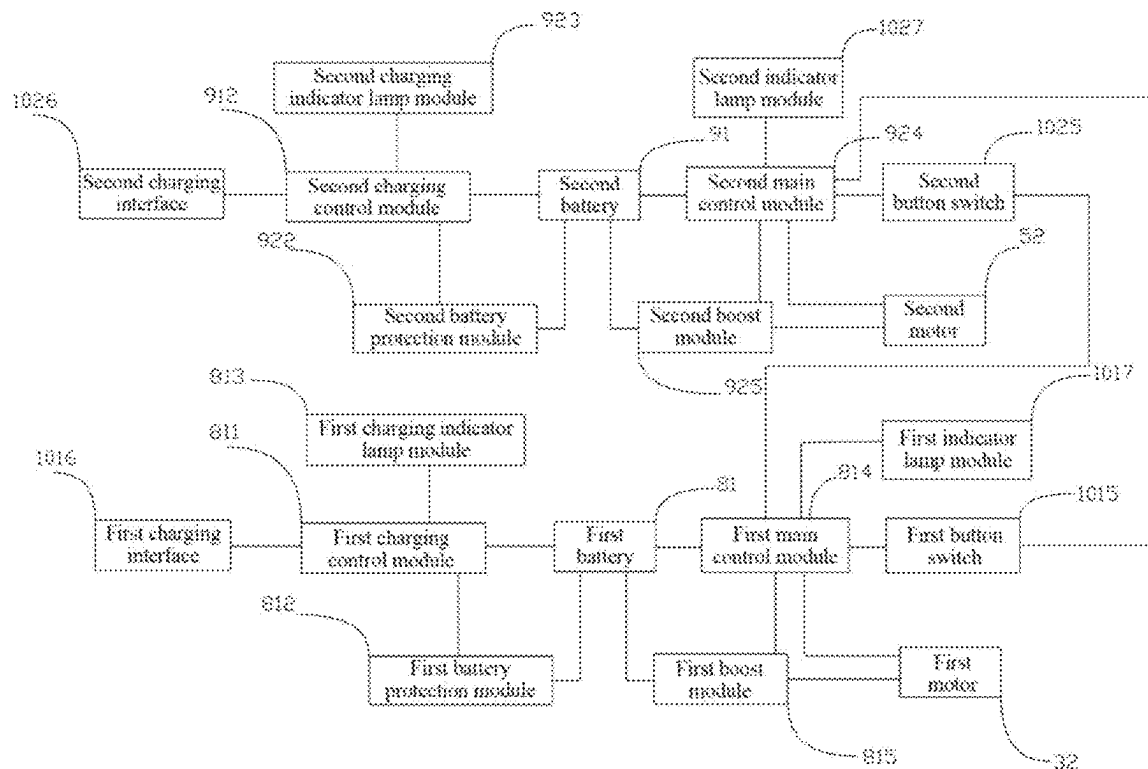


FIG. 7

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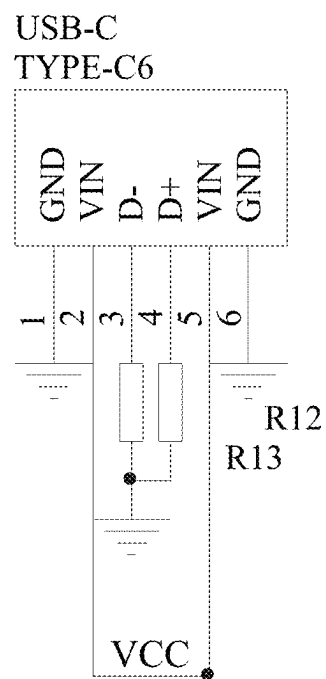


FIG. 8

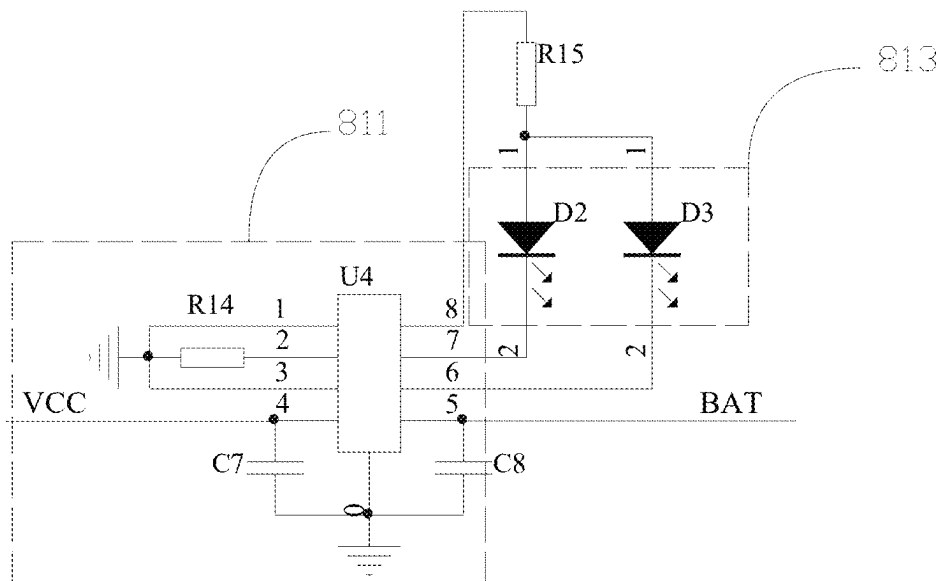


FIG. 9

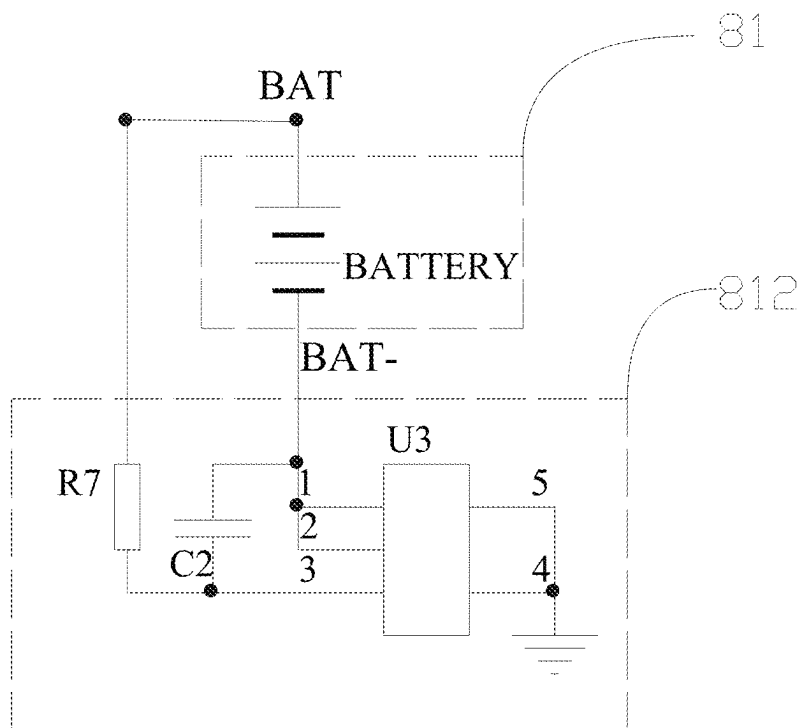


FIG. 10

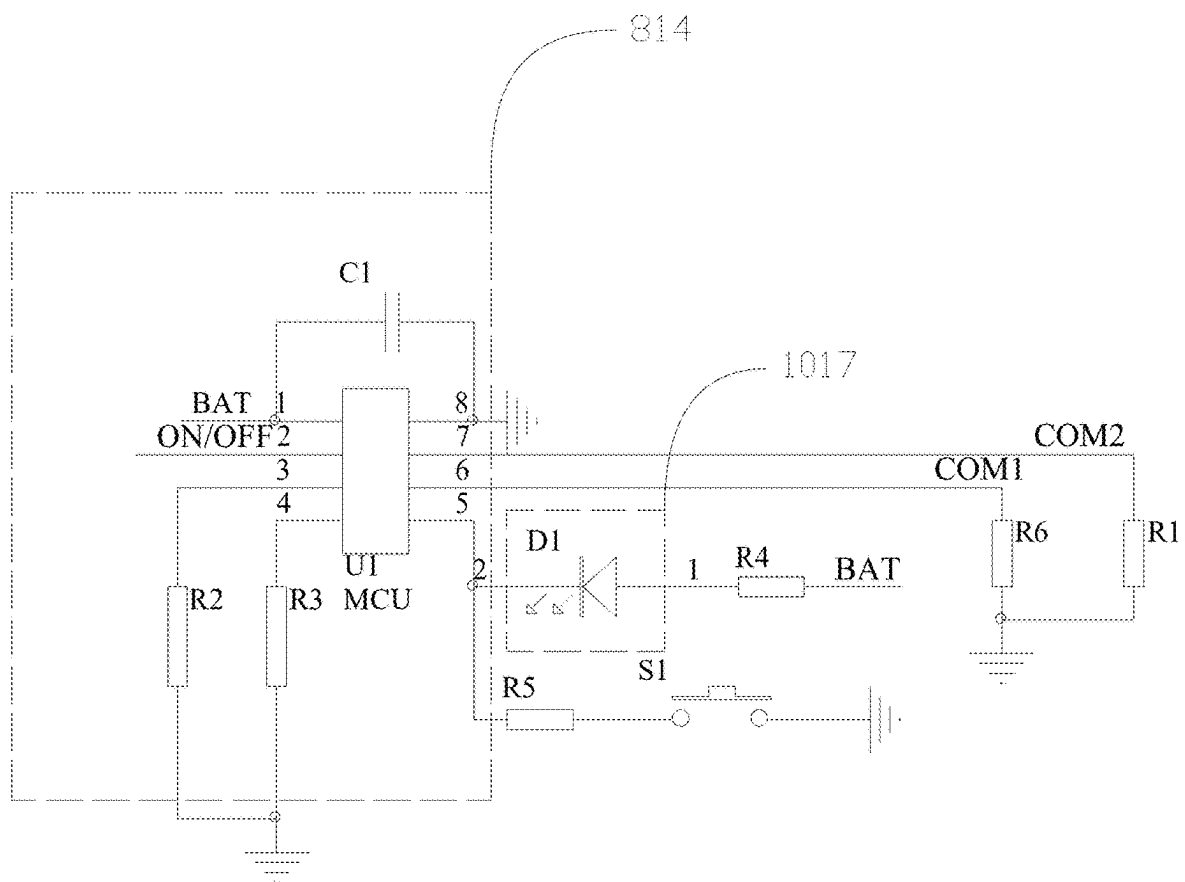


FIG. 11

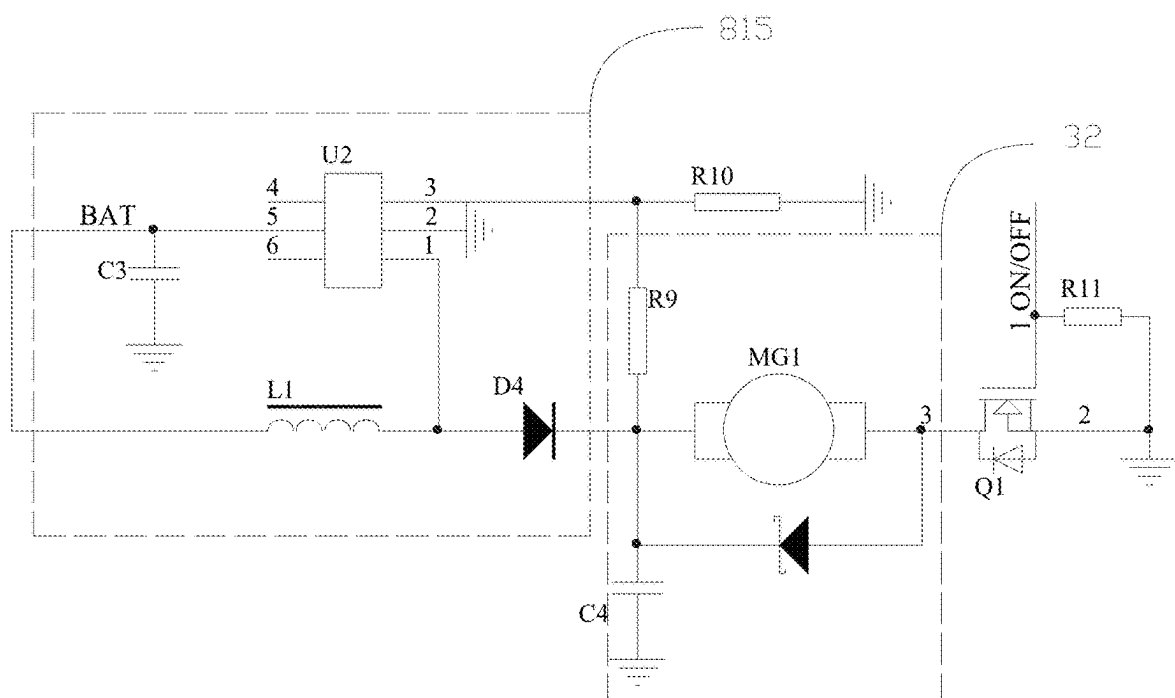


FIG. 12

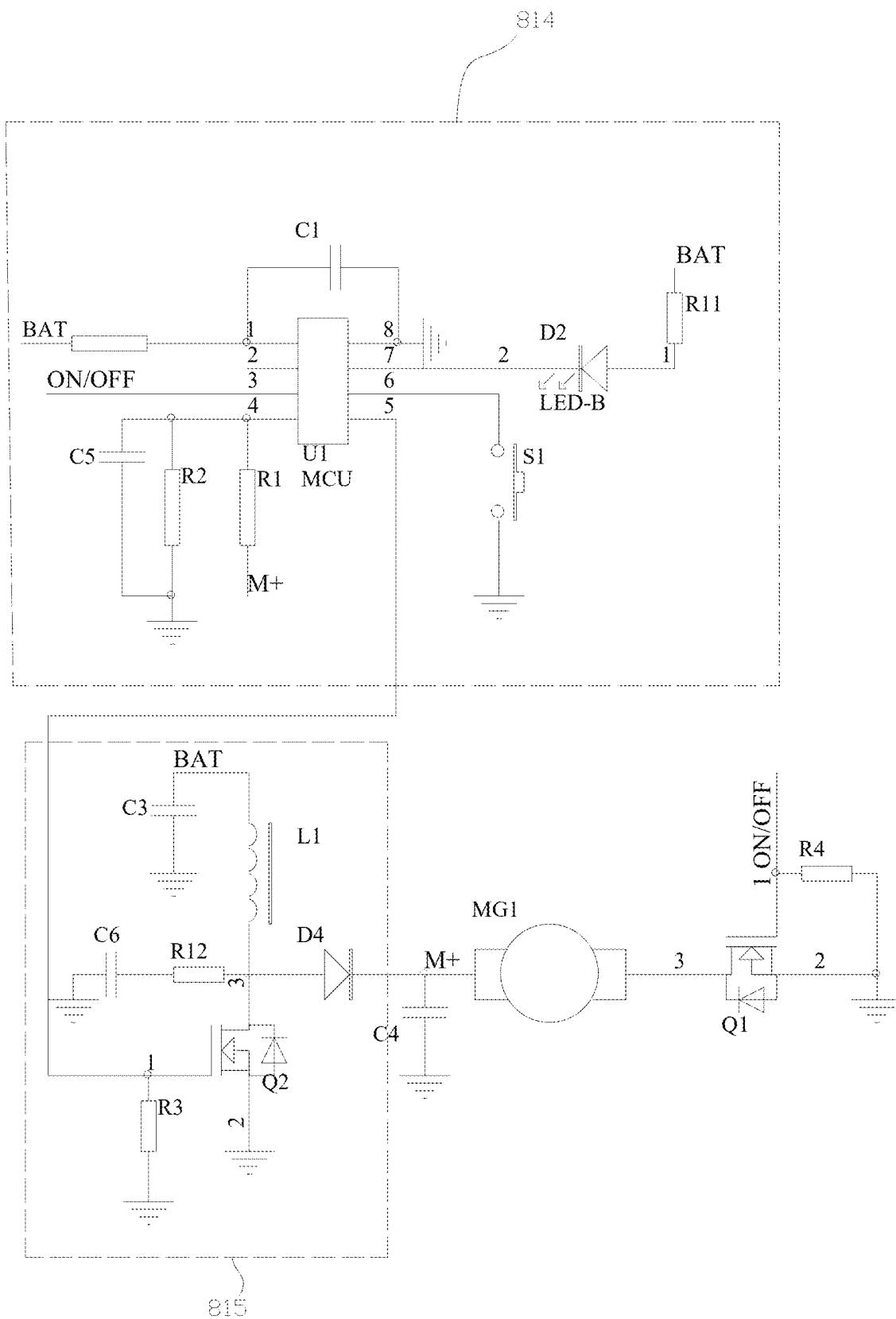


FIG. 13

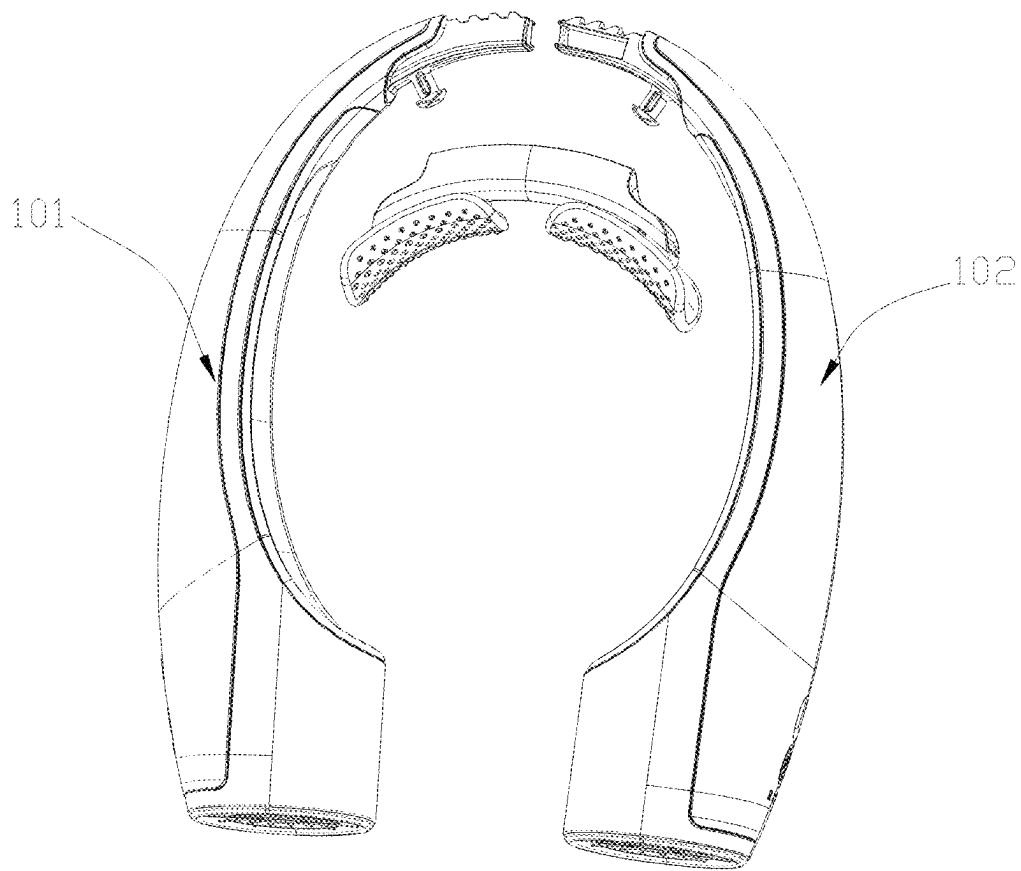


FIG. 14

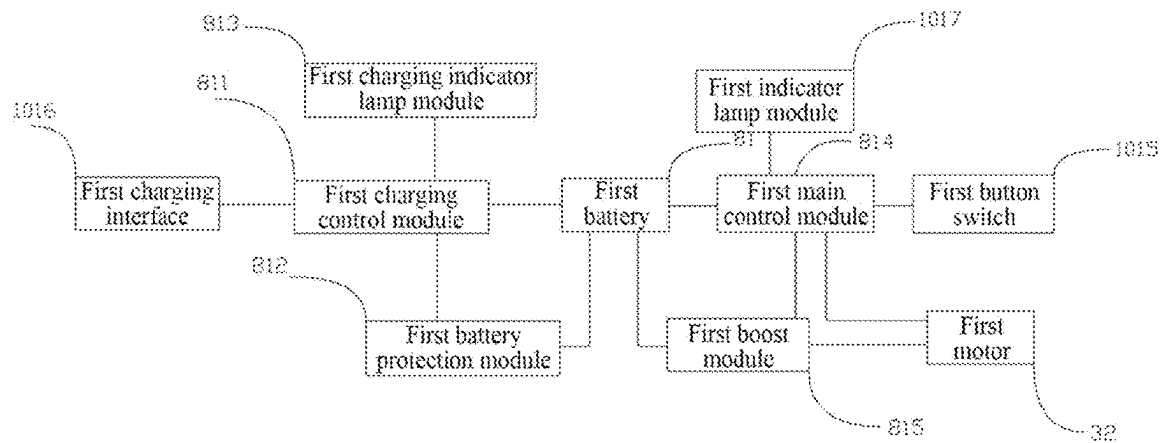


FIG. 15

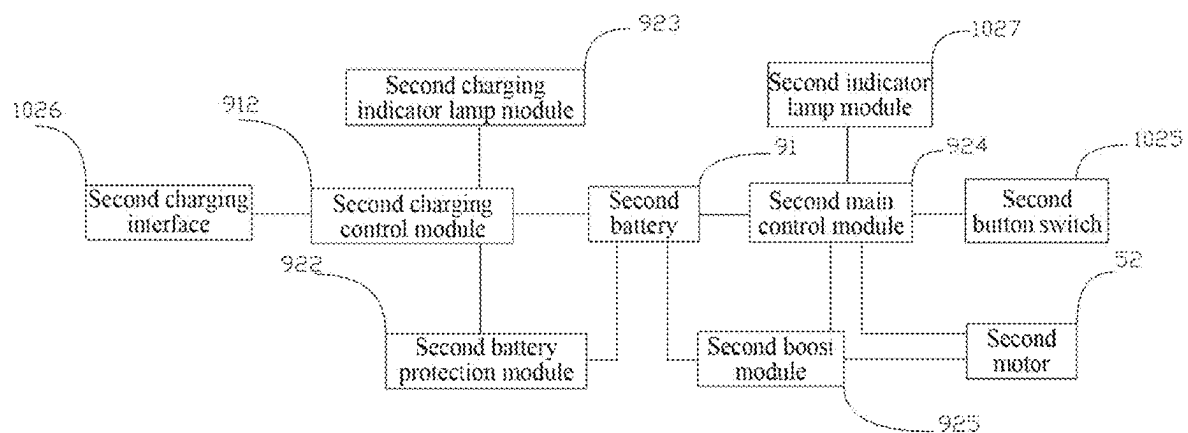


FIG. 16

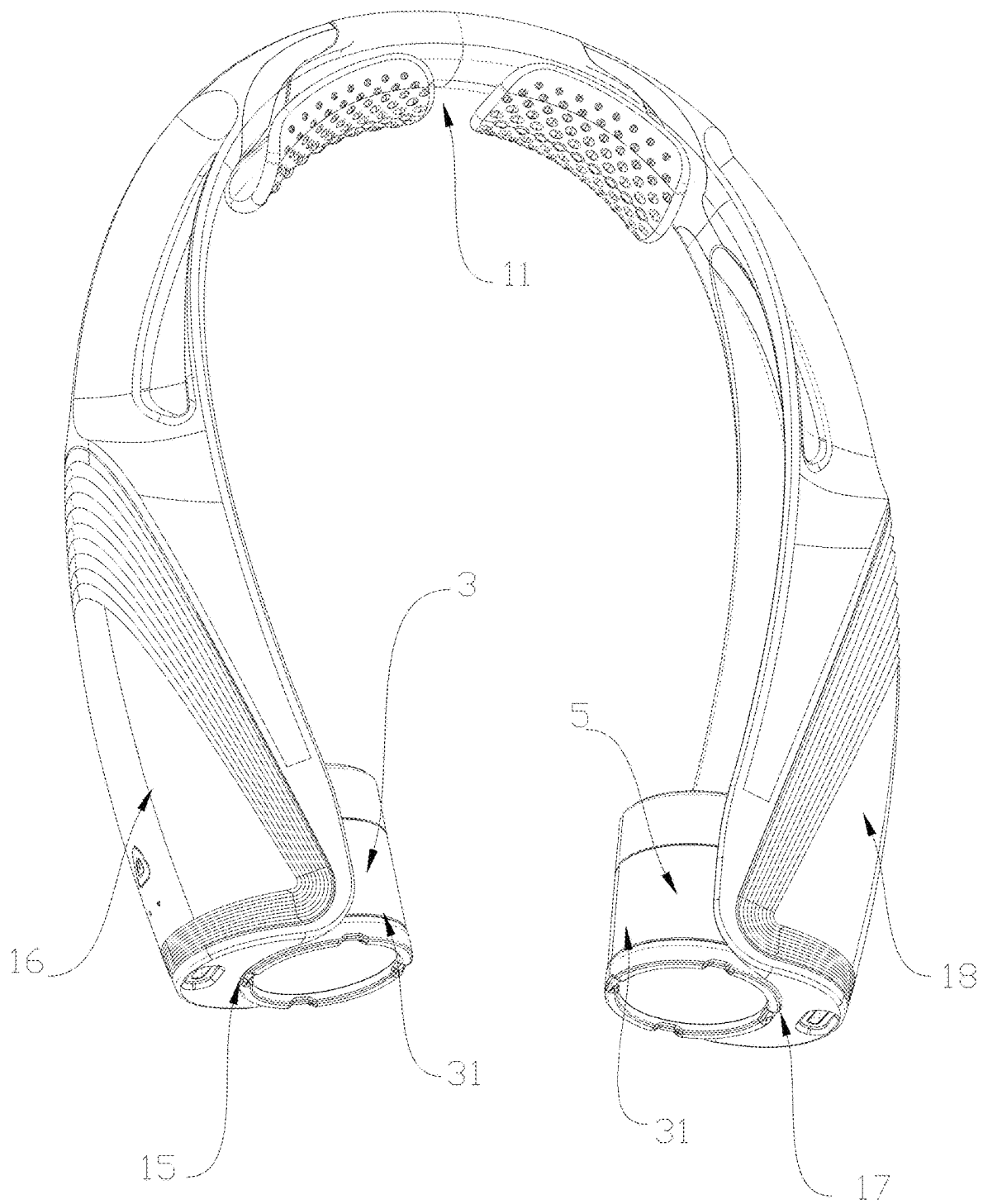


FIG. 17

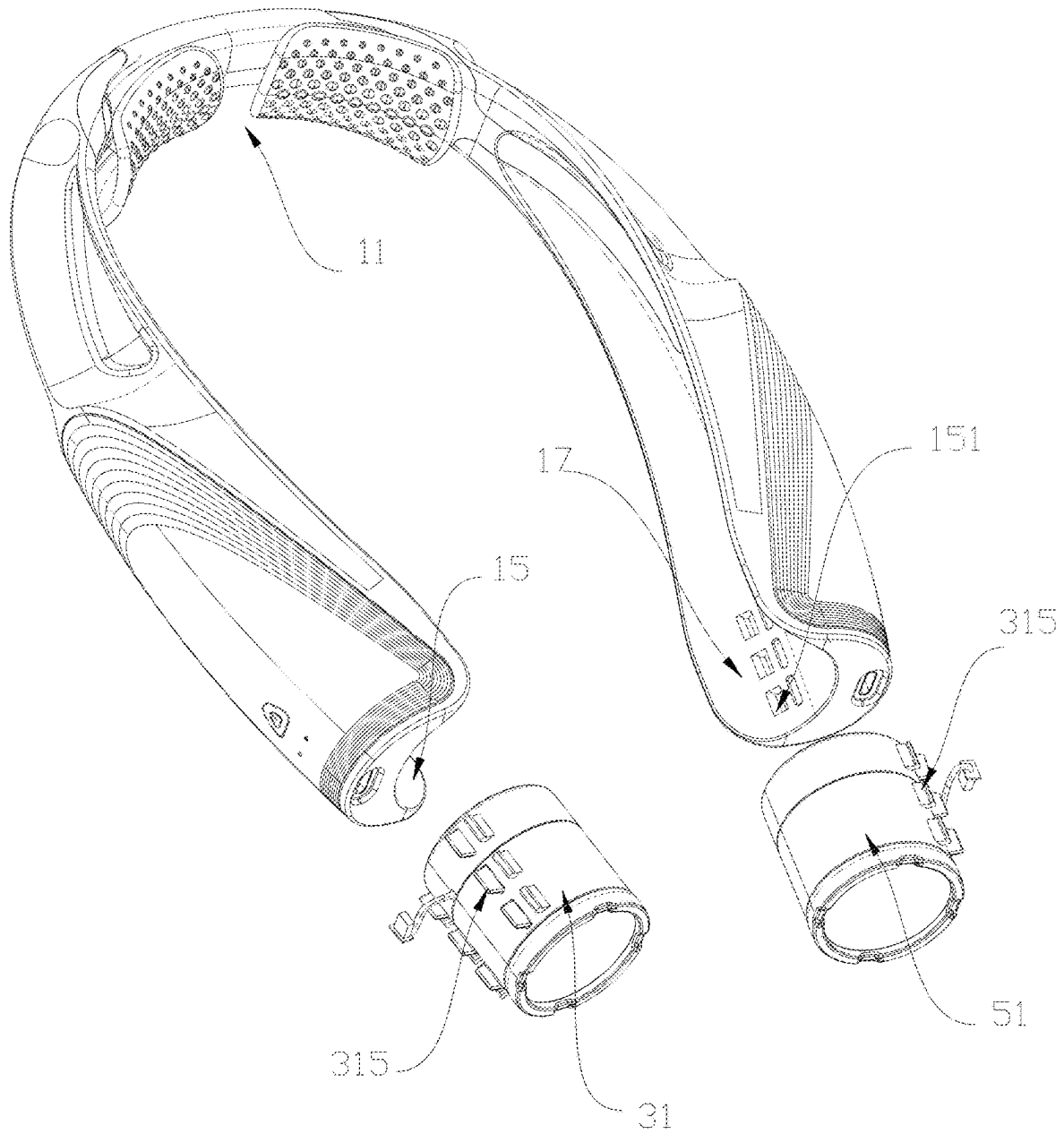


FIG. 18

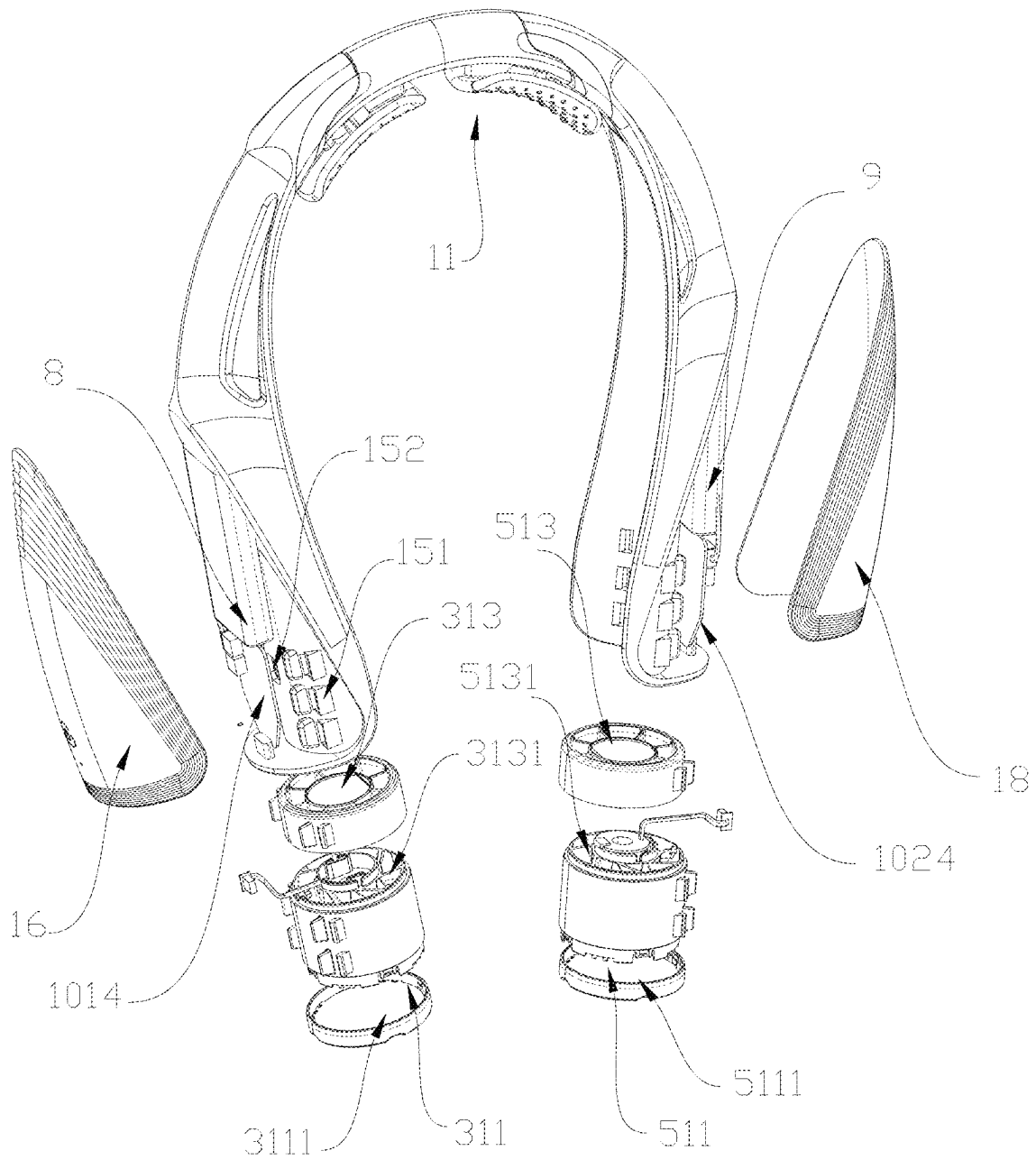


FIG. 19

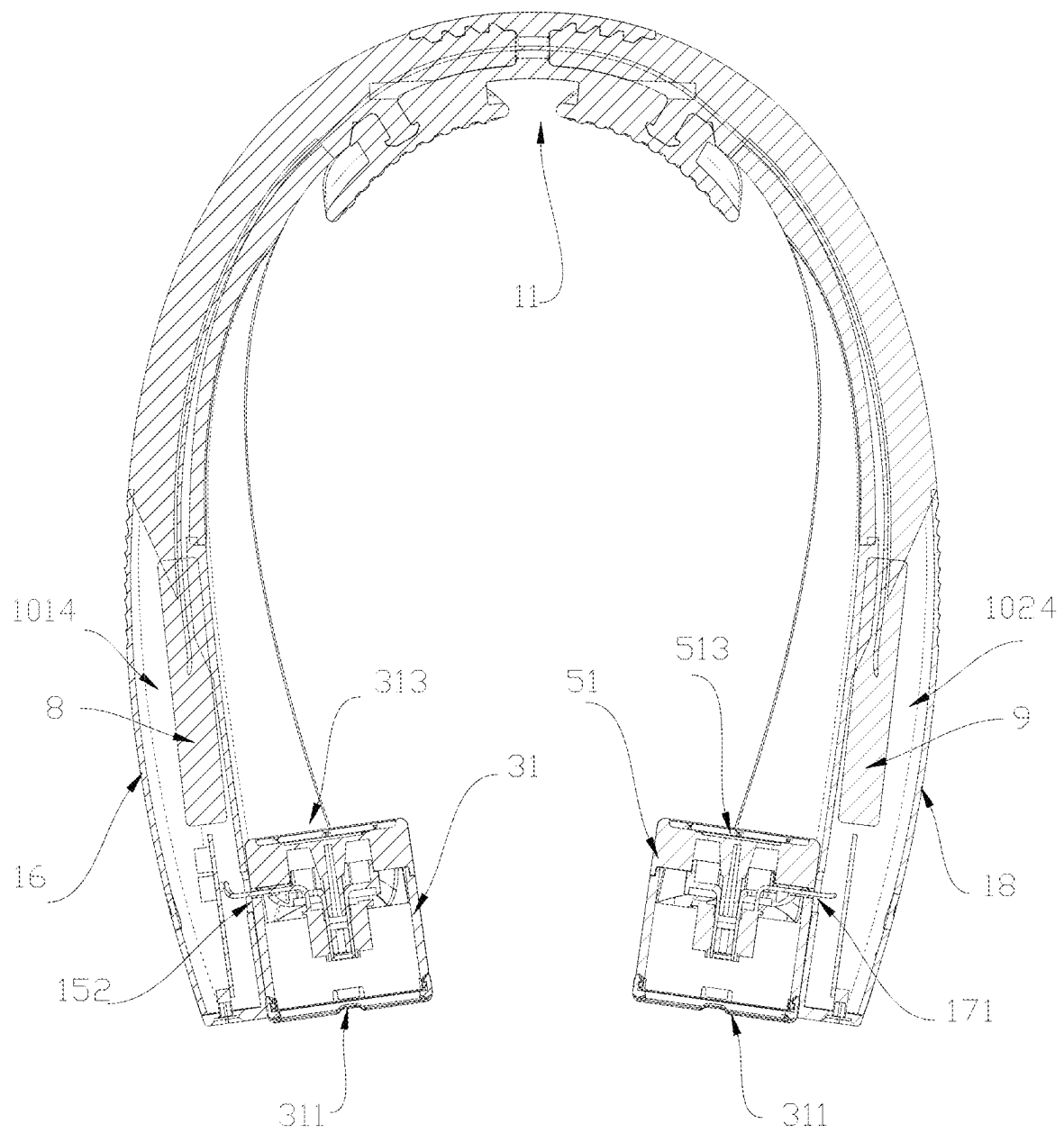


FIG. 20

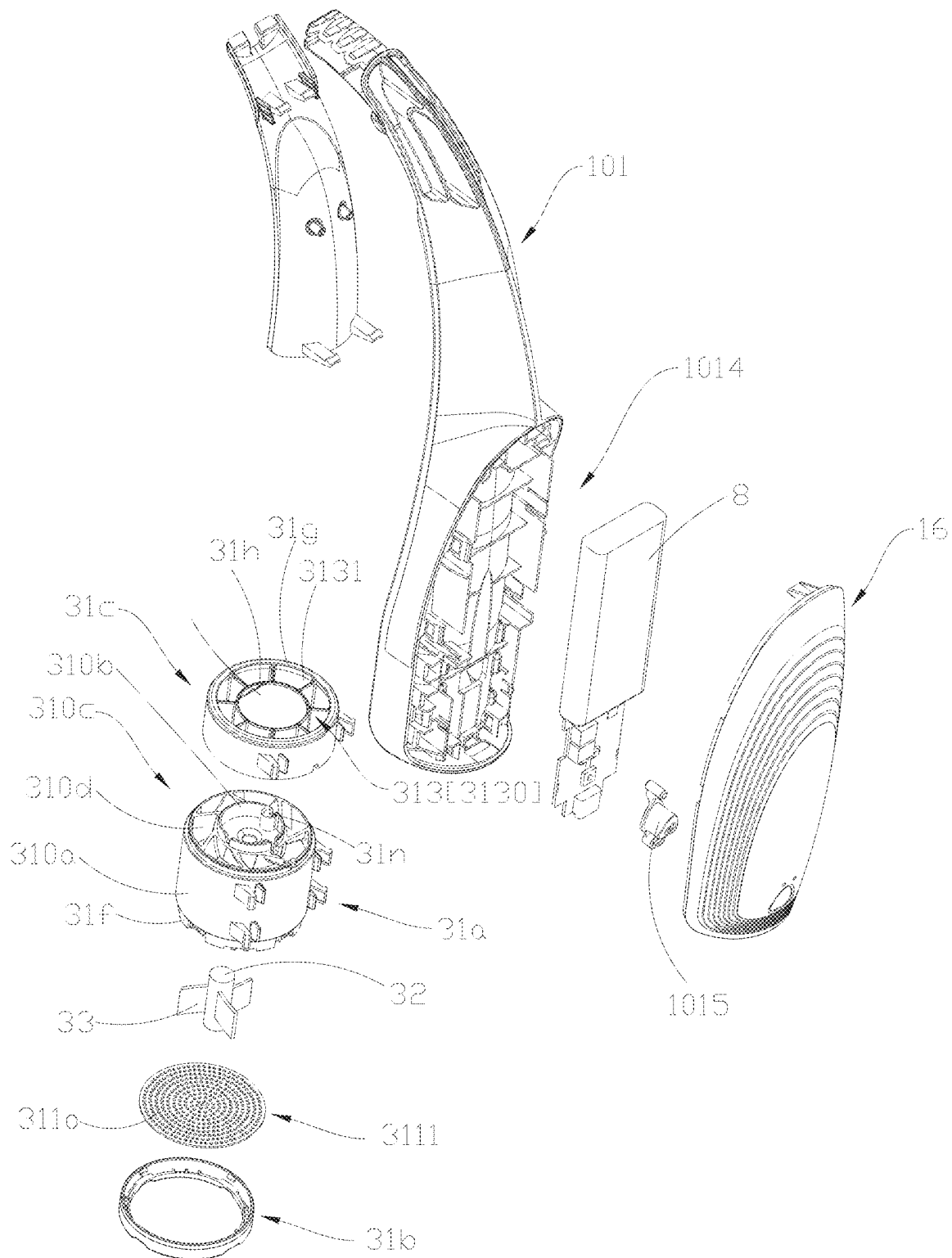


FIG. 21

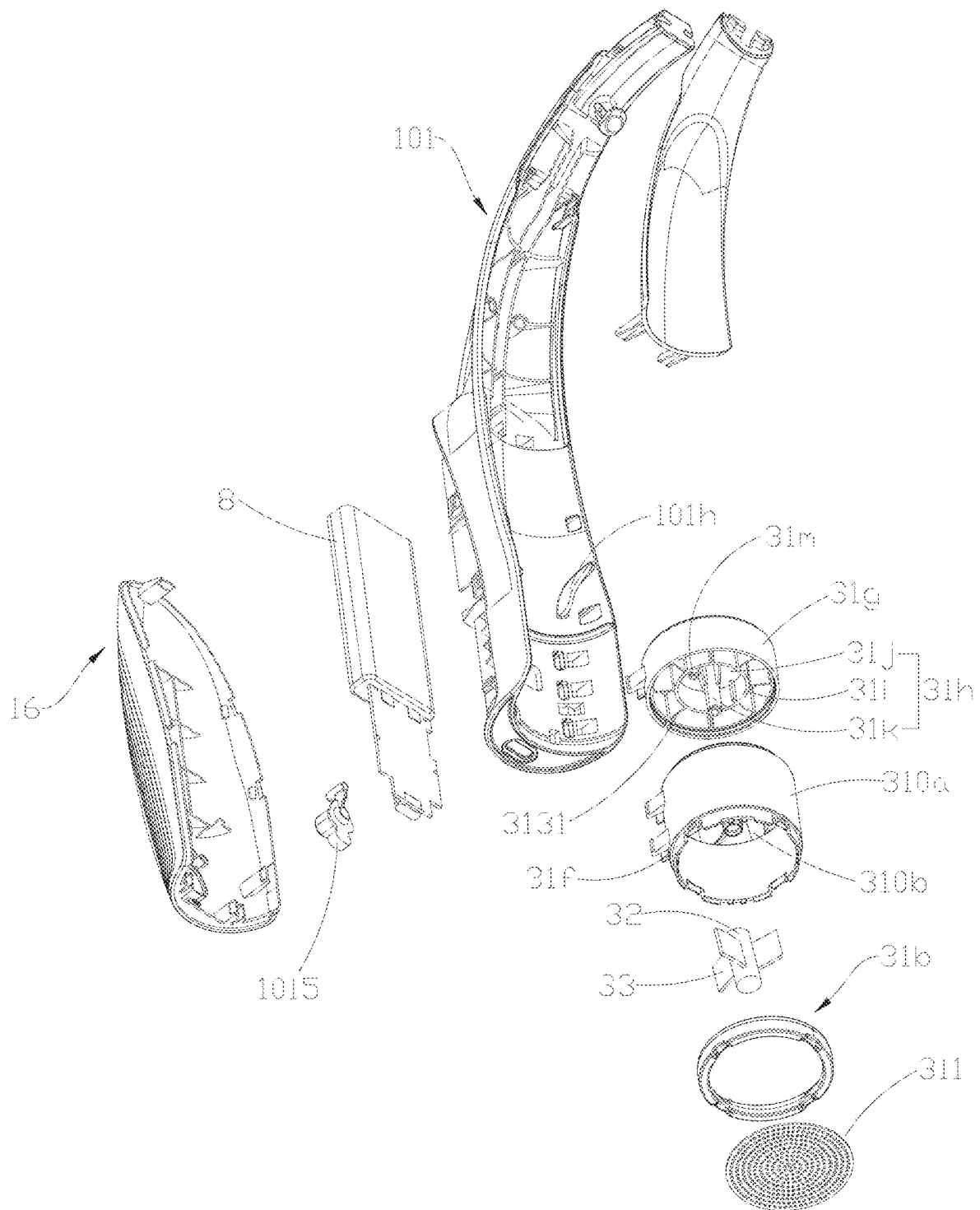


FIG. 22

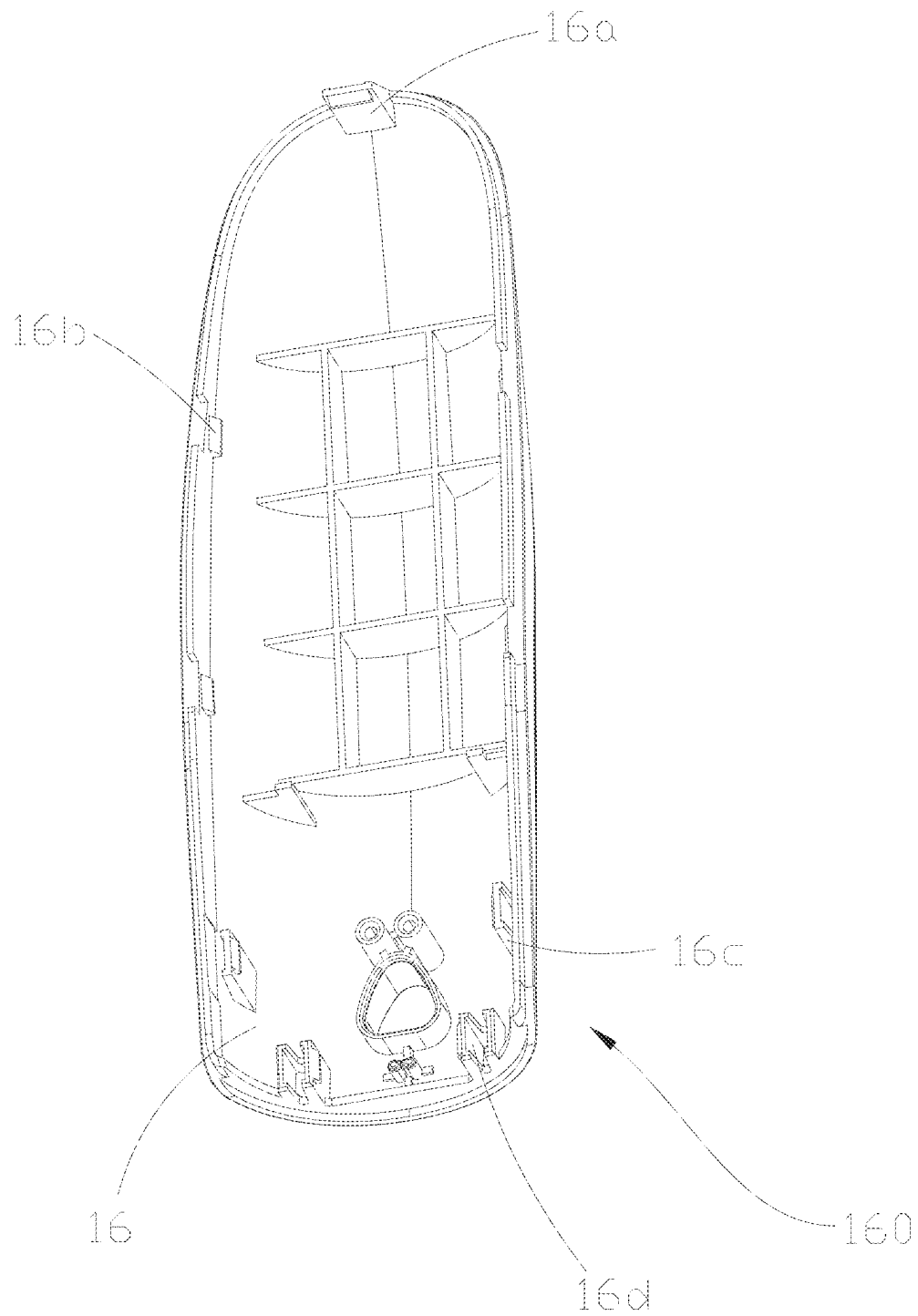


FIG. 23

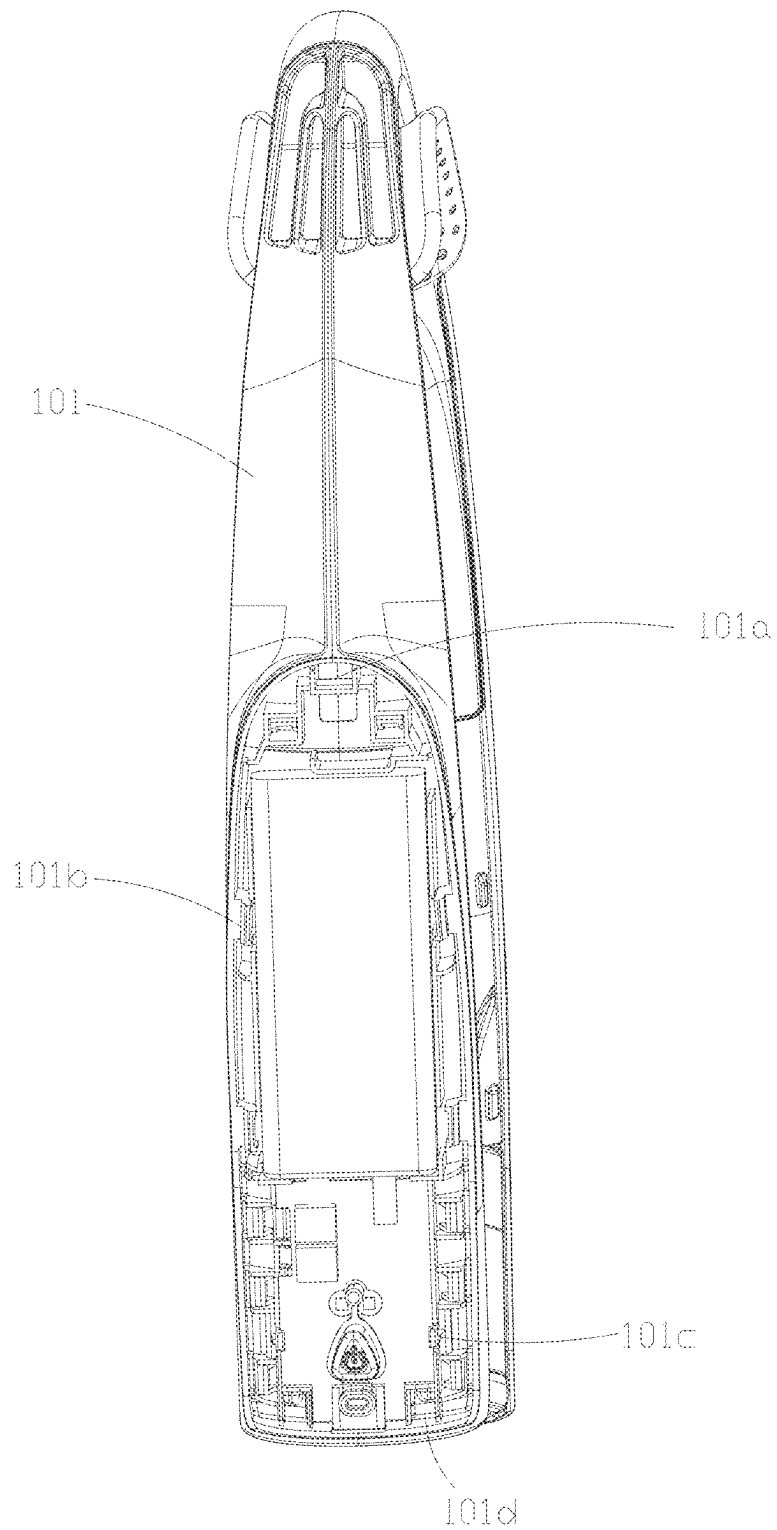


FIG. 24

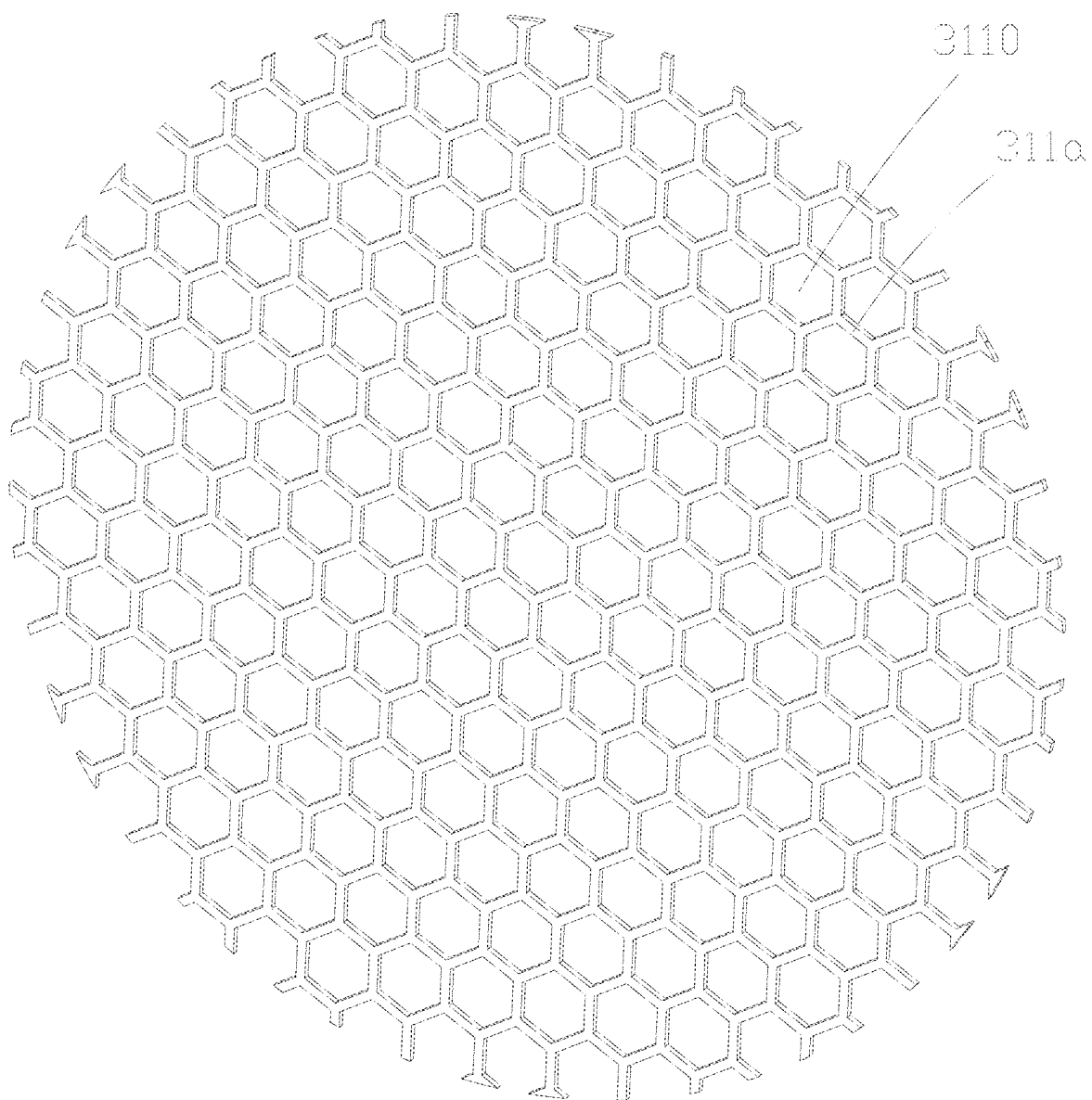


FIG. 25

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PORTABLE FAN NECK FAN AND FAN ASSEMBLY

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Continuation-in-Part of the U.S. application Ser., No. 18/933,625 filed on Oct. 31, 2024, and entitled "NECK FAN," now pending, which is a continuation-in-part of the U.S. application Ser., No. 18/421,213 filed on Jan. 24, 2024, the entire disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to the field of fans, and in particular, to a portable fan, a neck fan and a fan assembly.

BACKGROUND

As portable cooling equipment, a portable fan has received widespread attentions and been widely applied in hot seasons in recent years, especially a neck fan. The neck fan can be carried around and free the hands, and are favored by the majority of users.

However, as people continue to pursue comfortable experiences, they have gradually increasing expectations for the performance of the neck fan. In the scorching summer, users hope to achieve a faster and more direct cooling effect in a high-temperature environment, but the design of the traditional neck fan fails to fully meet this demand.

For this purpose, the present invention provides a neck fan that allows air from the fan to be blown more directly and centrally towards a user, thereby providing a more comfortable user experience in a hot environment. According to this design, the cooling effect of the neck fan is improved while the portability is achieved, so that the growing demand of the user for personalized and efficient cooling equipment is met.

SUMMARY

In order to overcome the shortcomings of the prior art, the present invention provides a portable fan, a neck fan and a fan assembly.

A portable fan includes a first main body portion and a first fan assembly. The first main body portion has an arc-shaped structure, and includes a concave first air guide path positioned on a surface of one side of the first main body portion and extended along an extending direction of the arc-shaped structure. The first fan assembly is arranged at one end of the first main body portion and is configured to blow air towards the first air guide path, and the first air guide path is configured to guide air of the first fan assembly to one side of the first main body portion and one end of the first main body portion away from the first fan assembly.

In one embodiment, one end of the portable fan away from the first fan assembly is further configured to connect the other portable fan through a connection member, so that the portable fan, the connection member and the other portable fan are able to be assembled into a neck fan for hanging around a user's neck, and the portable fan and the other portable fan are respectively configured to be arranged on each of the two sides of the user's neck.

In one embodiment, the first fan assembly includes a first fan shell, a first motor, and a first fan blade mounted on a rotating shaft of the first motor, the first motor and the first

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fan blade are arranged within the first fan shell; the first fan shell includes a shell main body and an air outlet hood, the air outlet hood is arranged on one side of the shell main body; the air outlet hood includes a plurality of air gathering rings and an annular wall structure connected to a periphery of the plurality of air gathering rings, a plurality of air outlet holes are formed among the plurality of air gathering rings, and the air gathering rings are configured to cooperate with an inner side surface of the annular wall structure to pressurize wind from the shell main body and then blow it out.

In one embodiment, the inner side surface of the annular wall structure is an inclined surface or an arc-shaped surface, so that an inner diameter of the annular wall structure gradually decreases along a direction from the shell main body to the air outlet hood.

In one embodiment, the air outlet hood further includes a mounting structure connected to the plurality of air gathering rings, and the mounting structure includes an annular mounting wall and a cover plate connected to the annular mounting wall.

In one embodiment, the mounting structure further includes a mounting column connected to an inner side of the cover plate, and the mounting column is detachably connected to the shell main body; the cover plate is in a circular shape and is positioned in a middle of the plurality of air outlet holes, and a groove is arranged on an outer surface of the cover plate.

In one embodiment, the mounting structure is further provided with a first alignment structure, the shell main body has a second alignment structure, and the first alignment structure and the second alignment structure cooperate to enable an aligned assembly of the shell main body and the air outlet hood.

In one embodiment, the shell main body includes an annularly arranged sidewall structure, an mounting portion positioned at one end inside the sidewall structure and an air guiding portion connected between the mounting portion and the sidewall structure, the air guiding portion includes a plurality of air guiding blades arranged at intervals and is configured to guide wind from the first fan blade based on a preset direction, the first fan blade and the first motor are arranged within the sidewall structure and are positioned on one side of the mounting portion, and the first motor is mounted on the mounting portion.

In one embodiment, the mounting portion is provided with an opening, a conducting wire connected to the first motor extends through the opening, and then stretches into the first main body portion to be electrically connected with a first power supply assembly.

In one embodiment, the first fan shell further includes a metal barrier net, the metal barrier net is arranged on one side of the sidewall structure away from the air outlet hood, the metal barrier net is made of metal material and is provided with a plurality of air inlet holes, and an inner diameter of each the air inlet holes ranges from 0.5 mm to 4 mm.

In one embodiment, the air inlet hole is in a regular hexagon shape, a surrounding wall is arranged between two adjacent air inlet holes, and a width of the surrounding wall ranges from 0.05 mm to 2 mm.

In one embodiment, the first fan shell further includes an air inlet cover, the metal barrier net is mounted on an inner side of the air inlet cover, the air inlet cover is arranged on one end of the shell main body away from the air outlet hood, the air inlet cover is in an annular shape, and the air inlet cover is sleeved on one end of the shell main body; an outer side surface of one end of the shell main body can be

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provided with a stepped groove, the air inlet cover is sleeved and mounted on the stepped groove, and an outer side surface of the air inlet cover is flush with the outer side surface of the shell main body.

In one embodiment, the first fan assembly is a vertical blowing type fan assembly with air inlet and air outlet positioned on the same axis, and the metal barrier net, the first fan blade, the first motor, the shell main body and the air outlet hood are all arranged in a straight line.

In one embodiment, the first main body portion is provided with a first mounting cavity, the portable fan further includes a first power supply assembly and a first outer cover, the first power supply assembly is positioned within the first mounting cavity, the first mounting cavity is provided with an opening, and the first outer cover is detachably covered on the opening of the first mounting cavity.

In one embodiment, an inner side of the first outer cover is provided with at least one first snap-fit member, an outer side of the first main body portion is provided with at least one second snap-fit member, and the at least one first snap-fit member and the at least one second snap-fit member are snap-fitted to enable an assembly of the first outer cover and the first main body portion.

In one embodiment, the first fan assembly is detachably arranged on an inner side of the first main body portion, a plurality of first latching members are arranged on the inner side of the first main body portion, and the first fan assembly is provided with a plurality of second latching members detachably clamped with the first latching members.

In one embodiment, the first main body portion further includes at least one air guiding vane, the at least one air guiding vane is arranged within the first air guide path adjacent to the first fan assembly, and is configured to guide the wind from the first fan assembly; the number of the at least one air guiding vane is two, the two air guiding vanes are respectively arranged on each of two sides of the first air guide path, and a distance between the two air guiding vanes gradually increases along an air outlet direction of the first air guide path.

The present invention also provides a neck fan, the neck fan includes the portable fan described above., the other portable fan and a connection member, the connection member is detachably connected to at least one of the first portable fan and the other portable fan assembly.

In one embodiment, the connection member is detachably connected to the portable fan and the other portable fan, the connection member includes a flexible connection portion, the flexible connection portion is provided with at least one mounting hole, the at least one mounting hole is configured to be sleeved and fixed onto a first connection end and a second connection end, so that first main body portion and a second main body portion are connected through the flexible connection portion; the at least one mounting hole is an mounting through hole arranged along an extending direction, each of two ends of the mounting hole is respectively configured to sleeve and fix the first connection end and the second connection end; the portable fan and the other portable fan are respectively provided with corresponding and button switches, so that the turning on and off of the portable fan and the other portable fan are independently controlled by the corresponding key switches.

The present invention also provides a fan assembly, the fan assembly includes a first fan shell, a first motor and a first fan blade connected to the first motor, the first fan blade is arranged within the first fan shell, the first fan shell includes a shell main body, an air outlet hood and a metal barrier net, the air outlet hood is arranged on one side of the shell main

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body and is provided with an air outlet hole, the metal barrier net is arranged on the other side of the shell main body, the air outlet hood includes a plurality of air gathering rings and an annular wall structure connected to a periphery of the plurality of air gathering rings, the metal barrier net includes a plurality of air inlet holes that can reduce wind resistance and increase air intake volume, the plurality of air outlet holes are defined among the plurality of air gathering rings, and the air gathering rings are configured to cooperate with an inner side surface of the annular wall structure to pressurize wind from the shell main body and then blow it out.

The beneficial effects of the present invention are as follows: the present invention provides a portable fan and a neck fan. By arranging the first air guide path, the wind of the first fan assembly can be guided to one side of the first main body portion and one end of the first main body portion away from the first fan assembly, thereby enlarging the blowing area of the first fan assembly and improving the user experience. Moreover, the fan assembly of the present invention is not only provided with an air outlet cover that can pressurize to improve the air outlet effect, but also is provided with a metal screen that can reduce wind resistance and increase the air intake volume, resulting in a relatively high wind efficiency.

BRIEF DESCRIPTION OF THE DRAWINGS

In order to explain the technical solutions of the embodiments of the present invention more clearly, the following will briefly introduce the accompanying drawings used in the embodiments. The drawings in the following description are only some embodiments of the present invention. Those of ordinary skill in the art can obtain other drawings based on these drawings without creative work.

The present invention is further described below in detail in combination with the accompanying drawings and embodiments.

FIG. 1 is a schematic diagram of an entire structure of a neck fan according to Embodiment I of the present invention;

FIG. 2 is a schematic diagram of an entire structure of another viewing angle of the neck fan according to Embodiment I of the present invention;

FIG. 3 is a sectional view of the neck fan according to Embodiment I of the present invention;

FIG. 4 is an exploded view of the neck fan according to Embodiment I of the present invention;

FIG. 5 is a partial enlarged structural diagram of a portion V of FIG. 3;

FIG. 6 is a partial enlarged structural diagram of a portion VI of FIG. 4;

FIG. 7 is a schematic block diagram of the neck fan according to Embodiment I of the present invention;

FIG. 8 is a circuit diagram of a first charging interface of the neck fan of the present invention;

FIG. 9 is a circuit diagram of a first charging control module of the neck fan of the present invention;

FIG. 10 is a circuit diagram of a first battery of the neck fan of the present invention;

FIG. 11 is a circuit diagram of a first main control module of the neck fan of the present invention;

FIG. 12 is a circuit diagram of a first boost module and a first motor of the neck fan of the present invention;

FIG. 13 is a circuit diagram of a first main control module, a first boost module and a first motor of the neck fan according to an alternative embodiment of the present invention;

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FIG. 14 is a schematic diagram of an entire structure of the neck fan according to Embodiment II of the present invention;

FIG. 15 is a schematic block diagram of a first main body portion of the portable fan according to Embodiment II of the present invention;

FIG. 16 is a schematic block diagram of a second main body portion of the portable fan according to Embodiment II of the present invention;

FIG. 17 is a schematic diagram of an entire structure of the neck fan according to Embodiment III of the present invention;

FIG. 18 is an exploded view of the neck fan according to Embodiment III of the present invention;

FIG. 19 is another exploded view of the neck fan according to Embodiment III of the present invention;

FIG. 20 is a cross-sectional view of the neck fan according to Embodiment III of the present invention;

FIG. 21 is an exploded view of a portable fan of the present invention;

FIG. 22 is an exploded view of FIG. 21 from another angle;

FIG. 23 is a schematic structural diagram of an outer side of a first main body portion of the portable fan shown in FIG. 21;

FIG. 24 is a schematic structural diagram of an inner side of a first outer cover body of the portable fan shown in FIG. 21; and

FIG. 25 is a schematic structural diagram of a metal screen of a first fan assembly of the portable fan according to another embodiment of the present invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

Embodiment I

Referring to FIG. 1 to FIG. 13, a neck fan includes a hanging main body 1 and a first fan assembly 3. A straight-tube-shaped first air guide chamber 2 is arranged at a lower end portion of the hanging main body 1 in a protruding manner; the first air guide chamber 2 includes a first air inlet 22 arranged at the lower end portion of the hanging main body 1 and a first air outlet 23 opposite to the first air inlet 22; the hanging main body 1 includes an inner end portion 11 that is in contact with the neck of a user; the first air outlet 23 faces the inner end portion 11; the first fan assembly 3 is located in the first air guide chamber 2; and the first fan assembly 3 is configured to drive an air flow to flow from the first air inlet 22 to the inner end portion 11 via the first air outlet 23. Through the above structure, due to the straight-tube shape of the first air guide chamber 2, the neck fan forms a vertical blowing type fan capable of blowing air in a vertical direction. The first fan assembly 3 can drive the air flow to be blown from the first air inlet 22 to the inner end portion 11 through the first air outlet 23, greatly improving the air blowing efficiency. A user can directly feel the cool air blown directly to the neck when using the neck fan. This improves the overall performance of the fan. Furthermore, since the first air guide chamber 2 is a straight-tube-shaped cavity integrally protruding out of the hanging main body 1, this design improves the overall sense of beauty and sense of design of the product, helps to improve the wearing comfort of the neck fan, and improves the overall quality and user experience of the product.

In this embodiment, a hanging main body 1 is U-shaped, the hanging main body 1 includes a first main body portion

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101, a second main body portion 102 and a connection member 103. The first main body portion 101 and the second main body portion 102 are detachably connected through the connection member 103. Furthermore, the connection member 103, which is flexible, can be a silica gel connection member, a plastic connection member or a rubber connection member. Moreover, the first fan assembly 3 is arranged on the first main body portion 101.

In this embodiment, the neck fan (such as the first main body portion 101) is provided with a first air guide path 12 that is inwards sunken; the first air guide path 12 extends from the first air outlet 23 to the inner end portion 11; and the first fan assembly 3 drives the air flow to be blown out from the first air outlet 23 towards the inner end portion 11 along the first air guide path 12. Further, a flexible first neck support 111 is connected to the inner end portion 11; a first air inlet gap 121 is formed between the first neck support 111 and the inner end portion 11; the first fan assembly 3 drives the air flow to be blown out from the first air outlet 23 towards the first air inlet gap 121 along the first air guide path 12. The first neck support 111 can be a silica gel neck support, a plastic neck support, a rubber neck support, or the like. Through the above structure, when the air flow is blown out from the first air outlet 23, under the guidance of the first air guide path 12 that is inwards sunken, the air flow will be blown towards the inner end portion 11 along the first air guide path 12. Due to the first air inlet gap 121, the air flow can be blocked in the gap, so that when a user wears the neck fan, the user can have a stronger feeling about the air flow at the inner end portion 11, and the heat dissipation effect is enhanced; and furthermore, the flexible neck support provides a more comfortable supporting effect for the user, which improves the user experience.

In this embodiment, a width of the first air guide path gradually decreases towards the inner end portion along the first air outlet; a width range of the first air guide path is 1 to 5 cm; a length range of the first air guide path is 8 to 16 cm; and a depth range of the first air guide path is 0 to 3 cm.

In this embodiment, the first main body portion 101 a first outer shell 1011 and a first inner shell 1012; the first air guide path 12 is arranged on the first inner shell 1012; and the first inner shell 1012 is detachably connected to the first outer shell 1011 through a buckle. It is convenient for a user to open and mount the first main body portion 101 for maintenance.

In this embodiment, a diameter of the first air inlet 22 is within a range of 30 mm to 60 mm, and a diameter of the first air outlet 23 is within a range of 30 mm to 60 mm. Through the above structure, an appropriate diameter range helps to achieve a balanced air flowrate, so that the fan can provide a sufficient air flow and maintain a good cooling effect. Furthermore, it helps to rationalize the overall structure, and the neck fan is convenient to carry and use.

In this embodiment, a straight-tube-shaped second air guide chamber 4 is further arranged at a lower end portion of the neck fan in a protruding manner; the second air guide chamber 4 includes a second air inlet 42 arranged at the lower end portion of the hanging main body 1 and a second air outlet 43 opposite to the second air inlet 42; the second air outlet 43 faces the inner end portion 11; the neck fan also includes a second fan assembly 5. The second fan assembly 5 is arranged in the second air guide chamber 4; and the second fan assembly 5 is configured to drive the air flow to be blown out from the second air inlet 42 towards the inner end portion 11 via the second air outlet 43. Through the above structure, due to the straight-tube shape of the second air guide chamber 4, the neck fan forms a vertical blowing

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type fan capable of blowing air in a vertical direction, so that the second fan assembly 5 can drive the air flow to be blown from the second air inlet 42 to the inner end portion 11 through the second air outlet 43, greatly improving the air blowing efficiency. A user can directly feel the cool air when using the neck fan. This improves the overall performance of the fan. Furthermore, since the second air guide chamber 4 is a straight-tube-shaped cavity integrally protruding out of the hanging main body 1, this design improves the overall sense of beauty and sense of design of the product, helps to improve the wearing comfort of the neck fan, and improves the overall quality and user experience of the product. It can be understood that The second fan assembly 5 is arranged in the second main body portion 102.

In this embodiment, the neck fan (such as the second main body portion 102) is provided with a second air guide path 13 that is inwards sunken; the second air guide path 13 extends from the second air outlet 43 to the inner end portion 11; and the second fan assembly 5 drives the air flow to be blown out from the second air outlet 43 towards the inner end portion 11 along the second air guide path 13. Further, a flexible second neck support 112 is connected to the inner end portion 11; a second air inlet gap 131 is formed between the second neck support 112 and the inner end portion 11; the second fan assembly 5 drives the air flow to be blown out from the second air outlet 43 towards the second air inlet gap 131 along the second air guide path 13. The second neck support 112 can be a silica gel neck support, a plastic neck support, a rubber neck support, or the like. Through the above structure, when the air flow is blown out from the second air outlet 43, under the guidance of the second air guide path 13 that is inwards sunken, the air flow will be blown towards the inner end portion 11 along the second air guide path 13. Due to the second air inlet gap 131, the air flow can be blocked in the gap, so that when a user wears the neck fan, the user can have a stronger feeling about the air flow at the inner end portion 11 more strongly, and the heat dissipation effect is enhanced; and furthermore, the flexible neck support provides a more comfortable supporting effect for the user, which improves the user experience.

In this embodiment, the second main body portion 102 includes a second outer shell 1021 and a second inner shell 1022; the second air guide path 13 is arranged on the second inner shell 1022; and the second inner shell 1022 is detachably connected to the second outer shell 1021 through a buckle. It is convenient for a user to open and mount the second main body portion 102 for maintenance. In particular, the first air guide path 12 includes a first air blowing opening 122 aligned with the first air outlet 23, and the second air guide path 13 includes a second air blowing opening 132 aligned with the second air outlet 43.

In this embodiment, the neck fan further includes a first power supply assembly 8; the first power supply assembly 8 is located inside the hanging main body 1; the first power supply assembly 8 is electrically connected to the first fan assembly 3 to supply power to the first fan assembly 3; the first power supply assembly 8 includes a first battery 81 and a first circuit board 82; and the first battery 81 is electrically connected to the first circuit board 82 to supply power to the first circuit board 82. The neck fan further includes a second power supply assembly 9; the second power supply assembly 9 is located inside the hanging main body 1; the second power supply assembly 9 is electrically connected to the second fan assembly 5 to provide power to the second fan assembly 5; the second power supply assembly 9 includes a second battery 91 and a second circuit board 92; and the second battery 91 is electrically connected to the second

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circuit board 92 to supply power to the second circuit board 92. Through the above structure, the first power supply assembly 8 is arranged inside the hanging main body 1. This design can improve the overall appearance of the neck fan and effectively supply power to the first fan assembly 3, to ensure the reliability of the product. The second power supply assembly 9 is arranged inside the hanging main body 1. This design can improve the overall appearance of the neck fan and effectively supply power to the second fan assembly 5, to ensure the reliability of the product.

In this embodiment, the first fan assembly 3 includes a first fan shell 31, a first motor 32 fixedly mounted inside the first fan shell 31, and a first fan blade 33 mounted on a rotating shaft of the first motor 32; the first fan shell 31 is provided with a third air inlet 311, a first air duct 312, and a third air outlet 313; and the first air inlet 22, the third air inlet 311, the first air duct 312, the third air outlet 313, the first air guide chamber 2, and the first air outlet 23 are connected to each other in sequence. Through the above structural design, the first fan assembly 3 in this embodiment is provided with the independent outer shell, which facilitates independent mounting and replacement of the fan.

In another implementation, the first fan assembly 3 includes a first motor 32 and a first fan blade 33 mounted on a rotating shaft of the first motor 32; the first air inlet 22 is connected with a first air inlet hood 221; the first air inlet hood 221 is provided with several first air inlet holes 222; the first air outlet 23 is connected to a first air outlet hood 231; and the first air outlet hood 231 is provided with several first air outlet holes 232. Through the above structure, the design of a fan shell is omitted in this case, which saves the cost. Furthermore, a direction of the air flow is effectively achieved. The design of the first air inlet hood 221 and the first air outlet hood 231 is conducive to preventing other debris, such as hairs and paper scraps, from being sucked into the fan, which is conducive to maintaining the safety of the fan.

In this embodiment, the second fan assembly 5 includes a second fan shell 51, a second motor 52 fixedly mounted inside the second fan shell 51, and a second fan blade 53 mounted on a rotating shaft of the second motor 52; the second fan shell 51 is provided with a fourth air inlet 511, a second air duct 512, and a fourth air outlet 513; and the second air inlet 42, the fourth air inlet 511, the second air duct 512, the fourth air outlet 513, the second air guide chamber 4, and the first air outlet 23 are connected to each other in sequence. Through the above structural design, the first fan assembly 3 in this embodiment is provided with the independent outer shell, which facilitates independent mounting and replacement of the fan.

In some other embodiments, the second fan assembly 5 includes a second motor 52 and a second fan blade 53 mounted on a rotating shaft of the second motor 52; the second air inlet 42 and a fourth air inlet 511 are connected with a second air inlet hood 421; the second air inlet hood 421 is provided with several second air inlet holes 422; the second air outlet 43 and a fourth air outlet 513 are connected with a second air outlet hood 431; and the second air outlet hood 431 is provided with several second air outlet holes 432. Through the above structure, the design of a fan shell is omitted in this case, which saves the cost. Furthermore, a direction of the air flow is effectively achieved. The design of the second air inlet hood 421 and the second air outlet hood 431 is conducive to preventing other debris, such as hairs and paper scraps, from being sucked into the fan, which is conducive to maintaining the safety of the fan.

In this embodiment, the connector **103** is provided with an installation hole **1031**; the first main body portion **101** is provided with a first connection end **1013** connected to the connector **103**; the first connection end **1013** is detachably inserted into one end of the installation hole **1031**; the second main body portion **102** is provided with a second connection end **1023** connected to the connector **103**; and the second connection end **1023** is detachably inserted into the other end of the installation hole **1031**.

Further, a concave-convex first buckle **10131** is arranged at the first connection end **1013**; the connector **103** is provided with a first buckle slot **1032** that cooperates with the first buckle **10131**; a concave-convex second buckle **10331** is arranged at the second connection end **1023**; and the connector **103** is provided with a second buckle slot **10231** that cooperates with the second buckle **10331**. Through the above structure, the connector **103** is a flexible silica gel connector. After the connector **103** is bent to an angle, the first main body portion **101** and the second main body portion **102** can be easily inserted into the installation hole **1031**. The first main body portion **101** is in clamping fit with the first buckle slot **1032** on an inner wall of the connector **103** through the concave-convex first buckle **10131** arranged at the first connection end **1013**, and the second main body portion **102** is in clamping fit with the second buckle slot **10231** on the inner wall of the connector **103** through the concave-convex second buckle **10331** arranged at the second connection end **1023**, so that it is convenient to mount and connect the first main body portion **101** to the second main body portion **102** to form a complete neck fan.

Referring to FIG. 3 to FIG. 6, the connection member **103** can be divided into a flexible connection portion **103a**, and the flexible connection portion **103a** can be provided with at least one installation hole **1031**. The at least one installation hole **1031** is configured to extend along an extending direction D1 from the first connection end **1013** of the first main body portion **101** to the second connection end **1023** of the second main body portion **102**, and the at least one installation hole **1031** is further configured to be sleeved and fixed onto the first connection end **1013** and the second connection end **1023**, so that the first main body portion **101** and the second main body portion **102** are connected through the flexible connection portion **103a**. The first main body portion **101** and the second main body portion **102** are connected through the flexible connection portion **103a**, so that a flexible connection is formed between the first main body portion **101** and the second main body portion **102**, thereby facilitating the wearing by users.

In this embodiment, the flexible connection portion **103a** is provided with a through installation through hole which is arranged along the extending direction D1, each of two ends of the installation hole **1031** is respectively configured to sleeve and fix the first connection end **1013** and the second connection end **1023**. Adopting one installation through hole is simple in process and offers flexible assembly space. However, in an alternative embodiment, the flexible connection portion **103a** may also have two blind holes (i.e., non-through holes) which are arranged oppositely. The two blind holes are respectively configured to sleeve and fix the first connection end **1013** and the second connection end **1023**.

In an embodiment, an inner sidewall of the at least one installation hole **1031** is provided with a first anti-detachment structure (such as a first buckle slot) and a second anti-detachment structure (such as a second buckle slot), the first anti-detachment structure is configured to be snap-fitted

and connected to the first buckle **10131** on an outer surface of the first connection end **1013**, the second anti-detachment structure is configured to be snap-fitted and connected to the second buckle **10331** on an outer surface of the second connection end **1023**, and at least one of the first anti-detachment structure and the first buckle **10131** includes a plurality of convex-concave structures arranged in sequence along the extending direction D1.

Specifically, in this embodiment, the first anti-detachment structure includes a plurality of first buckle slots **1023a** arranged along the extending direction, and the first buckle slot **1023a** is configured to be meshed and connected to a plurality of projections arranged along the extending direction on the first buckle **10131**; the second anti-detachment structure includes a plurality of second buckle slots **1023b** arranged along the extending direction, and the second buckle slot **1023b** is configured to be meshed and connected to a plurality of projections arranged along the extending direction on the second buckle **10331**; the first anti-detachment structure is configured to be detachably connected to the first buckle **10131**, and the second anti-detachment structure is configured to be detachably connected to the second buckle **10331**.

In an embodiment, the flexible connection portion **103a** is in an arc-shaped and provided with an inner side adjacent to a human neck and an outer side away from the human neck, the first anti-detachment structure is arranged on an inner sidewall of one side of the at least one installation hole **1031** adjacent to the outer side, and the second anti-detachment structure is arranged on an inner sidewall of the other side of the at least one installation hole **1031** adjacent to the outer side.

In an embodiment, the flexible connection portion **103a** is provided with a first fixing structure **103b** and a second fixing structure **103c**, the first fixing structure **103b** and the second fixing structure **103c** are respectively arranged on each of two ends of the flexible connection portion **103a**, the first fixing structure **103b** is configured to connect to a third fixing structure **1013a** of the first connection end **1013**, the second fixing structure **103c** is configured to connect to a fourth fixing structure **1023a** of the second connection end **1023**; the first fixing structure **103b** is configured to detachably connect to the third fixing structure **1013a**, and the second fixing structure **103c** is configured to detachably connect to the fourth fixing structure **1023a**.

In an embodiment, one of the first fixing structure **103b** and the third fixing structure **1013a** is a first fixing post **141**, one of the first fixing structure **103b** and the third fixing structure **1013a** is a first fixing hole **142**; one of the second fixing structure **103c** and the fourth fixing structure **1023a** is a second fixing post **143**, one of the second fixing structure **103c** and the fourth fixing structure **1023a** is a second fixing hole **144**, the first fixing structure **103b** extends along a direction D2 of a first tangent line perpendicular to the extending direction, and the second fixing structure **103c** extends along a direction D3 of a second tangent line perpendicular to the extending direction.

In an embodiment, the first fixing post **141** includes a first post body **1411** and a first buckle hook portion **1412** connected to one end of the first post body **1411** away from the flexible connection portion **103a**, and an outer diameter of the first buckle hook portion **1412** is greater than an outer diameter of the first post body **1411** to prevent the first fixing post **141** from detaching from the first fixing hole **142**; the second fixing post **143** includes a second post body **1431** and a second buckle hook portion **1432** connected to one end of the second post body **1431** away from the flexible connec-

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tion portion 103a, and an outer diameter of the second buckle hook portion 1432 is greater than an outer diameter of the second post body 1431 to prevent the second fixing post 143 from detaching from the second fixing hole 144.

In an embodiment, the first fixing hole 142 includes a first portion 1421 corresponding to the first post body 1411 and a second portion 1422 corresponding to the first buckle hook portion 1412, and a hole diameter of the second portion 1422 is greater than a hole diameter of the first portion 1421; the second fixing hole 144 includes a third portion 1441 corresponding to the second post body 1431 and a fourth portion 1442 corresponding to the second buckle hook portion 1432, and a hole diameter of the fourth portion 1442 is greater than a hole diameter of the third portion 1441.

In an embodiment, the connection member 103 includes at least one neck support (such as a first neck support portion 111 and a second neck support portion 112), the at least one neck support is connected to the inner side of the flexible connection portion 103a adjacent to the human neck, and the at least one neck support is configured to form at least one air inlet gap with the first main body portion 101 or the second main body portion 102, so that air of the first main body portion 101 or the second main body portion 102 can enter the air inlet gap (such as 121, 131). Thereby preventing the human neck from directly contacting with the first main body portion 101 or the second main body portion 102, and avoiding the situation that wind cannot be provided to a surface of the neck skin.

As shown previously, in this embodiment, the at least one neck support includes the first neck support portion 111 and the second neck support portion 112, the first neck support portion 111 is positioned adjacent to the first main body portion 101 and is configured to form a first air inlet gap 121 with the first main body portion 101, and the second neck support portion 112 is positioned adjacent to the second main body portion 102 and is configured to form a second air inlet gap 131 with the second main body portion 102.

In an embodiment, a plurality of projections 103m are arranged on surfaces of the first neck support portion 111 and the second neck support portion 112 away from the flexible connection portion 103a, and the plurality of projections 103m are configured to form a gap between the at least one neck support and the human neck skin. The plurality of projections 103m are further configured to avoid the situation that the human neck directly contacts the first main body portion 101 or the second main body portion 102, resulting in the inability to supply wind to the surface of the neck skin.

It can be seen that the surfaces of the first neck support portion 111 and the second neck support portion 112 away from the flexible connection portion 103a are arc-shaped. A size of the plurality of projections gradually decreases from a middle to two ends along a vertical direction D4 perpendicular to the extending direction D1. Thereby achieving a comfortable wearing effect.

In this embodiment, the flexible connection portion 103a, the first neck support portion 111 and the second neck support portion 112 are made of a flexible material, such as the same flexible material (such as a flexible silica gel material), and they are integrally formed. Thereby achieving a comfortable wearing effect.

Furthermore, the flexible connection portion 103a includes a main body portion 103d, a first extending portion 103e and a second extending portion 103f, the main body portion 103d is provided with the at least one installation hole 1031, the first extending portion 103e and the second extending portion 103f are respectively connected to each of two ends of the main body portion 103d, the first extending

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portion 103e and the second extending portion 103f protrude from the main body portion 103d along the extending direction D1, the first extending portion 103e and the second extending portion 103f are respectively configured to cover and connect to an inner surface of the first connection end 1013 adjacent to the human neck and an inner surface of the second connection end 1023 adjacent to the human neck. The first fixing structure and the second fixing structure are respectively arranged on the first extending portion 103e and the second extending portion 103f, thereby the first fixing structure and the second fixing structure being adjacent to an inner side of the human neck. The first anti-detachment structure and the second anti-detachment structure are positioned away from an outer side of the human neck, and the internal and external cooperation enables the connection member 103 to reliably connect the first main body portion 101 and the second main body portion 102, preventing the connection parts from easily getting detached.

In this embodiment, the first air guide chamber 2 is located inside the first main body portion 101; the first main body portion 101 is further provided with a first mounting chamber 1014; and the first power supply assembly 8 is located inside the first mounting chamber 1014. The second air guide chamber 4 is located inside the second main body portion 102; the second main body portion 102 is further provided with a second mounting chamber 1024; and the second power supply assembly 9 is located inside the second mounting chamber 1024.

In this embodiment, the first main body portion 101 is further provided with a first button switch 1015, a first charging interface 1016, and a first indicator lamp module 1017; and the first button switch 1015, the first charging interface 1016, and the first indicator lamp module 1017 are all electrically connected to the first power supply assembly 8. The second main body portion 102 is further provided with a second button switch 1025, a second charging interface 1026, and a second indicator lamp module 1027; and the second button switch 1025, the second charging interface 1026, and the second indicator lamp module 1027 are all electrically connected to the second power supply assembly 9. Through the above structure, both the first button switch 1015 and the second button switch 1025 are configured to turn on or turn off the neck fan. The first charging interface 1016 and the second charging interface 1026 are configured to charge the first battery 81 and the second battery 91, respectively. The first indicator lamp module 1017 and the second indicator lamp module 1027 can be configured to: display whether the fan is in an ON or OFF state and display a wind rating of the fan.

In this embodiment, referring to FIG. 7 to FIG. 13, the neck fan further includes a first charging control module 811, a first battery protection module 812, a first charging indicator module 813, a first main control module 814, and a first boost module 815 which are electrically connected to the first circuit board 82.

The first charging interface 1016 and the first charging control module 811 are both electrically connected to the first battery 81; the first charging indicator module 813 is connected to the first charging control module 811; the first charging interface 1016 and the first charging control module 811 are configured to charge the first battery 81; the first charging indicator module 813 is configured to display a state of charging; and the first battery protection module 812 is electrically connected to the first charging control module 811 and the first battery 81 to protect the first battery 81.

The first main control module 814, the first boost module 815, and a first motor 32 are all electrically connected to the

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first battery **81**; the first button switch **1015** and the first indicator lamp module **1017** are electrically connected to the first main control module **814**; and the first indicator lamp module **1017** is configured to display turning on or turning off of the neck fan.

In this embodiment, the neck fan further includes a second charging control module **921**, a second battery protection module **922**, a second charging indicator module **923**, a second main control module **924**, and a second boost module **925** which are electrically connected to the second circuit board **92**.

The second charging interface **1026** and the second charging control module **921** are both electrically connected to the second battery **91**; the second charging indicator module **923** is connected to the second charging control module **921**; the second charging interface **1026** and second first charging control module **921** are configured to charge the second battery **91**; the second charging indicator lamp module **923** is configured to display a state of charging; and the second battery protection module **922** is electrically connected to the second charging control module **921** and the second battery **91** to protect the second battery **91**.

The second main control module **924**, the first boost module **925**, and a second motor **52** are all electrically connected to the second battery **91**; the second button switch **1025** and the second indicator lamp module **1027** are electrically connected to the second main control module **924**; and the second indicator lamp module **1027** is configured to display turning on or turning off of the neck fan.

In this embodiment, the first main body portion **101** is provided with a first conductive end **1018**; the second main body portion **102** is provided with a second conductive end **1019**; when the first conductive end **1018** is communicated to the second conductive end **1019**, the first button switch **1015** is electrically connected to the first main control module **814** and the second main control module **924** respectively, and the second button switch **1025** is electrically connected to the first main control module **814** and the second main control module **924** respectively; and the first button switch **1015** or the second button switch **1025** is configured to send a third fan working signal.

The first main control module **814** is configured to: receive the third fan working signal and drive the first boost module **815** to receive and boost the output voltage of the first battery **81**, to drive the first motor **32** to work; and

the second main control module **924** is configured to: receive the third fan working signal and drive the second boost module **925** to receive and boost the output voltage of the second battery **91**, to drive the second motor **52** to work.

In this embodiment, the first main body portion **101** and the second main body portion **102** can be regarded as two independent fans electrically connected to each other to form the neck fan. The circuit diagrams of various circuit modules included in the two independent fans are the same.

Through the above structure, the first main body portion **101** and the second main body portion **102** are combined and spliced to form the complete U-shaped neck fan. At this time, the first conductive end **1018** is in conductive connection with the second conductive end **1019**. When the user presses the first button switch **1015** or the second button switch **1025**, the third fan working signal can be sent. At this time, after receiving the third fan working signal, the first main control module **814** drives the first boost module **815** to receive and boost the output voltage of the first battery **81**, thereby driving the first motor **32** to work. The rotation of the first motor **32** can drive a first fan blade **33** to rotate, achieving the air blowing effect of the first fan assembly.

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After receiving the third fan working signal, the second main control module **924** drives the second boost module **925** to receive and boost the output voltage of the second battery **91**, thereby driving the second motor **52** to operate. The rotation of the second motor **52** can drive a second fan blade **53** to rotate, achieving the air blowing effect of the second fan assembly.

It can be understood, the first main control module **814** of FIG. **11** and the first boost module **815** may have other embodiments. Referring to FIG. **12**, in other embodiment, a boost control chip **U2** of the first boost module **815** can be omitted. Specially, in the embodiment of FIG. **13**, the first boost module **815** includes an inductor **L1** and a switch element **Q2**, a control terminal of an MCU of the first main control module **814** is electrically connected to a control terminal of the switch element **Q2**, a first connection terminal of the switch element **Q2** is electrically connected to the battery **BAT** via the first inductor **L1**, and a second connection terminal of the switch element **Q2** is grounded. The first connection terminal of the switch element **Q2** is also electrically connected to the first motor **MG1**, in detail, the first connection terminal of the switch element **Q2** is also electrically connected to the first motor **MG1** via a diode **D4**. The MCU of the first main control module **814** is configured to turn on/turn off the switch element **Q2**, such that a voltage of the battery **BAT** can boost, and a boosted voltage can be provided to the first motor **MG1** to drive the first fan blade **33**. Further, the second main control module **924** and the second boost module **925** of FIG. **5** also can have the same structure of the first boost module **815** and the first main control module **814** of FIG. **13**, such that another boosted voltage can be provided to the second motor to drive the second fan blade **53**.

Embodiment II

Referring to FIG. **14** to FIG. **16**, a difference between Embodiment II and Embodiment I is only that the first main body portion **101** and the second main body portion **102** of the neck fan are removed to form two independent fans, so that the first main body portion **101** and the second main body portion **102** are two small fans capable of working alone.

In this embodiment, the first main body portion **101** includes a first charging control module **811**, a first battery protection module **812**, a first charging indicator module **813**, a first main control module **814**, and a first boost module **815** which are electrically connected to the first circuit board **82**.

The first charging interface **1016**, the first charging control module **811**, and the first charging indicator lamp module **813** are all electrically connected to the first battery **81**; the first charging interface **1016** and the first charging control module **811** are configured to charge the first battery **81**; the first charging indicator lamp module **813** is configured to display a battery level of the first battery **81**; and the first battery protection module **812** is electrically connected to the first charging control module **811** and the first battery **81** to protect the first battery **81**.

The first main control module **814**, the first boost module **815**, and a first motor **32** are all electrically connected to the first battery **81**; the first button switch **1015** and the first indicator lamp module **1017** are electrically connected to the first main control module **814**; the first button switch **1015** is configured to send a first fan working signal; the first main control module **814** is configured to: receive the first fan working signal and drive the first boost module **815** to

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receive and boost an output voltage of the first battery **81**, to drive the first motor **32** to work; and the first indicator lamp module **1017** is configured to display turning on or turning off of the neck fan.

Through the above structure, the first charging interface **1016** is configured to charge the first battery **81**; the first charging control module **811** is configured to control the charging of the first battery **81**; when the first button switch **1015** is turned on, the first fan working signal is sent; when receiving the first fan working signal, the first main control module **814** is configured to drive the first boost module **815** to receive and boost the output voltage of the first battery **81**, to drive the first motor **32** to work; and the first main control module **814** can further adjust the speed of the first motor **32** by controlling the output voltage of the first boost module **815**.

In this embodiment, the second main body portion **102** includes a second charging control module **921**, a second battery protection module **922**, a second charging indicator module **923**, a second main control module **924**, and a second boost module **925** which are electrically connected to the second circuit board **92**.

The second charging interface **1026**, the second charging control module **921**, and the second charging indicator lamp module **923** are all electrically connected to the second battery **91**; the second charging interface **1026** and the second charging control module **921** are configured to charge the second battery **91**; the first charging indicator lamp module **923** is configured to display a battery level of the second battery **91**; and the second battery protection module **922** is electrically connected to the second charging control module **921** and the second battery **91** to protect the second battery **91**.

The second main control module **924**, the second boost module **925**, and a second motor **52** are all electrically connected to the second battery **91**; the second button switch **1025** and the second indicator lamp module **1027** are electrically connected to the second main control module **924**; the second button switch **1025** is configured to send a second fan working signal; the second main control module **924** is configured to: receive the second fan working signal and drive the second boost module **925** to receive and boost an output voltage of the second battery **91**, to drive the second motor **52** to work; and the second indicator lamp module **1027** is configured to display turning on or turning off of the neck fan.

Through the above structure, the second charging interface **1026** is configured to charge the second battery **91**; the second charging control module **921** is configured to control the charging of the second battery **91**; when the second button switch **1025** is turned on, the second fan working signal is sent; when receiving the second fan working signal, the second main control module **924** is configured to drive the second boost module **925** to receive and boost the output voltage of the second battery **91**, to drive the second motor **52** to work; and the second main control module **924** can further adjust the speed of the second motor **52** by controlling the output voltage of the second boost module **925**. This structural design achieves an effect of using independent portions as independent fans after the neck fan is disassembled, bringing higher flexibility, and a user can select an appropriate usage method according to a specific need.

Embodiment III

Referring to FIG. 17 to FIG. 24, Embodiment III of the present invention provides a neck fan. It can be understood

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that the descriptions of Embodiment I and Embodiment II mentioned above are basically applicable to the neck fan of Embodiment III. Moreover, the following mainly describes the key contents of Embodiment III and the difference from Embodiment I and Embodiment II.

The neck fan includes a hanging main body **1**. The hanging main body **1** includes a first main body portion **101**, a second main body portion **102** and a connection member **103** connected to the first main body portion **101** and the second main body portion **102**, a first fan assembly **3** arranged at one end of the first main body portion **101** away from the connection member **103**, and a second fan assembly **5** arranged at one end of the second main body portion **102** away from the connection member **103**.

The connection member **103** is detachably connected to both the first main body portion **101** and the second main body portion **102**; the first main body portion **101** is detachably connected to the first fan assembly **3**; and the second main body portion **102** is detachably connected to the second fan assembly **5**. Thus, all five portions of the neck fan are detachably connected, forming a modular design. This not only facilitates assembly but also reduces the costs of assembly and maintenance. Furthermore, users can disassemble, assemble, store by themselves and conduct DIY designs and so on, thereby improving the user experience.

The first main body portion **101** has an arc-shaped structure, and includes a concave first air guide path **12** positioned on a surface of one side of the first main body portion **101** and extended along an extending direction of the arc-shaped structure. Moreover, the first fan assembly **3** is arranged at one end of the first main body portion **101** and is configured to blow air towards the first air guide path **12**. The first air guide path is configured to guide air of the first fan assembly **3** to one side of the first main body portion **101** (one side of the user's neck) and one end of the first main body portion **101** away from the first fan assembly **3** (such as the back of the user's neck).

The second main body portion **102** has an arc-shaped structure, and includes a concave second air guide path **13** positioned on a surface of one side of the second main body portion **102** and extended along an extending direction of the arc-shaped structure. Moreover, the second fan assembly **5** is arranged at one end of the second main body portion **102** and is configured to blow air towards the second air guide path **13**. The second air guide path is configured to guide air of the second fan assembly **5** to one side of the second main body portion **102** (the other side of the user's neck) and one end of the second main body portion **102** away from the second fan assembly **5** (such as the back of the user's neck).

Furthermore, the first main body portion **101** and the first fan assembly **3** can also be assembled into one to form a first portable fan, and the second main body portion **102** and the second fan assembly **5** can also be assembled into one to form a second portable fan. Thus, a neck fan can be disassembled into two portable fans for use according to users' needs. It can be understood that when used as a neck fan, the portable fan and the other portable fan are respectively arranged on both sides of the user's neck. Moreover, the fact that a neck fan can be disassembled into two portable fans for use according to users' needs can enrich usage scenarios, facilitate storage and improve the user experience.

The first fan assembly **3** includes a straight-tube-shaped first fan shell **31**, the first fan shell **31** is detachably connected to the hanging main body **1** and is located within a

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first accommodating chamber **15**, and the first fan shell **31** is provided with a third air inlet **311** and a third air outlet **313**.

Through the above structure, due to the straight-tube shape of the first fan shell **31**, the neck fan is formed into a vertical blowing type fan capable of blowing air in a vertical direction. The first fan assembly **3** can drive the air flow to be blown from the third air inlet **311** towards the user's neck via the third air outlet **313** along the first air guide path **12**, greatly improving the air blowing efficiency. A user can directly feel the cool air blown directly to the neck when using the neck fan, thereby improving the overall performance of the fan. Furthermore, the first fan shell **31** is detachably connected and assembled with the hanging main body **1**, so that separate mold opening, machining, and assembling can be achieved during production. The production efficiency is improved; the production cost is reduced; Moreover, it is also convenient for storage and transportation; and the portability of the product is improved.

The first fan shell **31** can include a shell main body **31a**, an air inlet cover **31b**, a metal barrier net **3111** and an air outlet hood **31c**.

The first fan blade **33** and the first motor **32** are arranged within the shell main body **31a**. Specifically, the shell main body includes a cylindrical sidewall structure **310a**, an mounting portion **310b** positioned at one end inside the sidewall structure **310a** and an air guiding portion **310c** connected between the mounting portion **310b** and the sidewall structure **310a**. The first fan blade **33** and the first motor **32** are arranged within the sidewall structure **310a** and are positioned on one side of the mounting portion **310b**, and the first motor **32** is mounted on the mounting portion **310b**.

The air guiding portion **310c** includes a plurality of air guiding blades **310d** arranged at intervals and is configured to guide wind from the first fan blade **33** based on a preset direction. Specifically, the thickness of each air guiding blade **310d** can gradually increase along a direction from the first fan blade **33** to the third air outlet **313**, thus achieving a better air guiding effect. The air inlet cover **31b** can be an annular shaped, and the air inlet cover **31b** is sleeved on one end of the shell main body **31a**. Specifically, an outer side surface of one end of the shell main body **31a** can be provided with a stepped groove **31f**; the air inlet cover **31b** is sleeved and mounted on the stepped groove **31f**. After mounting, the outer side surface of the air inlet cover **31b** is flush with the outer side surface of the shell main body **31a**. The metal barrier net **3111** can be mounted on the inner side of the air inlet cover **31b**.

The metal barrier net **3111** is provided with the third air inlet **311**, the third air inlet **311** includes a plurality of air inlet holes. Each air inlet hole **311a** can be a circular shaped, and the diameters of the plurality of air inlet holes **311a** can be the same. The diameter of each the air inlet hole **311a** can range from 0.5 mm to 4 mm. Such a size can prevent hair or debris from being sucked into the interior of the first fan assembly **3**. It can be understood that the metal barrier net **3111** is relatively thin and lightweight, which is conducive to the miniaturization and lightweighting of the product. In addition, since it is made of metal material, the size of the air inlet holes of the metal barrier net **3111** can be arranged to be relatively small, thereby effectively preventing foreign objects from entering the first fan assembly **3**.

The third air outlet **313** includes a plurality of air outlet holes arranged on the air outlet hood **31c**. Specifically, the air outlet hood **31c** includes an annular wall structure **31g**, an mounting structure **31h** and a plurality of air gathering rings **3131**. The plurality of air gathering rings **3131** are

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connected between the annular wall structure **31g** and the mounting structure **31h**. Moreover, the plurality of air outlet holes are formed among the plurality of air gathering rings.

The inner side surface of the annular wall structure **31g** is an inclined surface or an arc-shaped surface, so that an inner diameter of the annular wall structure **31g** gradually decreases along a direction from the third air inlet **311** to the third air outlet **313**. Moreover, the blowing wind has a pressurizing effect and the outlet wind intensity is improved. The mounting structure **31h** includes an annular mounting wall **31i**, a cover plate **31j** connected to the annular mounting wall **31i** and an mounting column **31k** connected to an inner side of the cover plate **31j**, and the mounting column **31k** is connected to the mounting portion **310b**. Specifically, the mounting column can be detachably connected to an mounting hole of the mounting portion **310b**, so that a detachable connection is formed between the air outlet hood **31c** and the shell main body **31a**. Moreover, the cover plate **31j** is in a circular shape and is positioned in a middle of the plurality of air outlet holes **3130**, and a groove is arranged on an outer surface of the cover plate **31j** for converging the outlet air.

The mounting structure **31h** is further provided with a first alignment structure **31m**, the mounting portion **310b** has a second alignment structure **31n**, and the first alignment structure **31m** and the second alignment structure **31n** cooperate to enable an aligned assembly of the shell main body **31a** and the air outlet hood **31c**. The first alignment structure **31m** and the second alignment structure **31n** can be one, two or more. Specifically, one of the first alignment structure **31m** and the second alignment structure **31n** can be an alignment hole, and the other one can be an alignment column that cooperates with the alignment hole.

The metal barrier net **3111** can prevent hair or debris from being sucked into the first fan shell, thereby avoiding damage to the fan assembly and reducing the wind force of blowing of the fan. The air gathering ring **3131** can gather the air blown towards the neck, thereby improving the blowing performance.

The first main body portion **101** further includes at least one air guiding vane **101h**, the at least one air guiding vane **101h** is arranged within the first air guide path **12** adjacent to the first fan assembly **3**, and is configured to guide the wind from the first fan assembly **3**.

In this embodiment, the number of the at least one air guiding vane **101h** is two, the two air guiding vanes **101h** are respectively arranged on each of two sides of the first air guide path **12**. Moreover, a distance between the two air guiding vanes **101h** gradually increases along an air outlet direction of the first air guide path **12**.

In this embodiment, a first conductive through hole **152** is provided in an inner side of the hanging main body adjacent to the first accommodating chamber **15**. The first conductive through hole **152** is configured to allow a wire of the first fan assembly **3** to pass through to be communicated to a first power supply assembly **8**. Moreover, the mounting portion **310b** can be provided with an opening **310e**. The wire connected to the motor can pass through the opening **310e** and extend out, and then be connected to the first power supply assembly **8** via the first conductive through hole **152**.

Through the above structure, when the neck fan needs to be assembled, the first outer cover **16** is first opened, and then the wire of the first fan assembly **3** is communicated to the first power supply assembly **8** in the first mounting chamber **1014** through the first conductive through hole **152**. Then, the first outer cover **16** is closed. After the wire of the first fan assembly **3** is connected, the first fan shell is

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assembled with the hanging main body through the buckling between the connection buckles 315 and the connecting buckle slots 151, and finally a vertical blowing type neck fan that is conductive with the hanging main body 1 and easy to mount is obtained.

In this embodiment, a second accommodating chamber 17 is further inwards sunken in one side, opposite to the first accommodating chamber 15, of the lower end portion of the second main body portion 102 of the hanging main body. The neck fan further includes a second fan assembly 5. The second fan assembly 5 includes a straight-tube-shaped second fan shell 51. The second fan shell 51 is detachably connected to the hanging main body and is located in the second accommodating chamber 17. The second fan shell 51 is provided with a fourth air inlet 511 and a fourth air outlet 513. The first fan assembly 5 is configured to drive an air flow to be blown from the fourth air inlet 511 towards the inner end portion via the fourth air outlet 513. Through the above structure, similarly, due to the straight-tube shape of the second fan shell 51, the neck fan is formed into a vertical blowing type fan capable of blowing air in a vertical direction. The second fan assembly can drive the air flow to be blown from the fourth air inlet 511 towards the inner end portion 11 via the fourth air outlet 513, greatly improving the air blowing efficiency. A user can directly feel the cool air blown directly to the neck when using the neck fan, thereby improving the overall performance of the fan. Furthermore, the second fan shell 51 is detachably connected and assembled with the hanging main body 1, so that separate mold opening, machining, and assembling can be achieved during production. The production efficiency is improved; the production cost is reduced; it is also convenient for storage and transportation; and the portability of the product is improved.

In this embodiment, several connection buckle slots 151 are provided in an inner side of the second main body portion 102 of the hanging main body adjacent to the second accommodating chamber 17; several connection buckles 315 matched and clamped with the connection buckle slots 151 are arranged on an outer wall of the second fan shell 51; and the second fan shell 51 is in detachably buckled connection to the hanging main body 1. Specifically, the connection buckle slots 151 are symmetrically distributed on an inner side of the second main body portion 102 of the hanging main body. There are six groups of connection buckle slots 151. Each group includes two connection buckle slots 151. The connection buckles 315 are symmetrically located on the outer wall of the second fan shell. There are six groups of connection buckles 315, and each group includes two connection buckles 315.

In this embodiment, the second main body portion 102 of the hanging main body is detachably connected with a second outer cover 18 on an outer side away from the second accommodating chamber 17. As in Embodiment I, the second main body portion 102 of the hanging main body 1 is provided with a second mounting chamber 1024, and the second power supply assembly 9 is located inside the second mounting chamber 1024. The second mounting chamber 1024 has an opening, and the second outer cover 18 is detachably covered at the opening.

Further, the specific structure of the second fan assembly 5 can be the same as that of the first fan assembly 3. The second fan assembly 5 further includes a second motor 52 and a second fan blade 53 mounted on a rotating shaft of the second motor. The second fan shell 51 can be basically the same in structure as the first fan shell 31. For example, it can include a shell main body 31a, an air inlet cover 31b, a metal

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barrier net 3111 and an air outlet hood 31c. The fourth air inlet 511 can also include a plurality of air inlet holes 311a arranged on the metal barrier net 3111 of the second fan shell 51. The fourth air outlet can also include a plurality of air outlet holes 3130 arranged on the air outlet hood 31c of the second fan shell 51. The specific structure of the second main body portion 102 can be the same as that of the first main body portion 101, and the second main body portion 102 and the first main body portion 101 can be arranged symmetrically. The second fan assembly 5 and the first fan assembly 3 can also be arranged symmetrically. Therefore the specific structures of the second fan shell 51 and the second main body portion 102 will not be introduced repeatedly here.

In this embodiment, a second conductive through hole 171 is provided in an inner side of the hanging main body adjacent to the second accommodating chamber 17. The second conductive through hole 171 is configured to allow a wire of the second fan assembly 5 to pass through to be communicated to the second power supply assembly 9.

Through the above structure, after the assembling of the first fan shell 31 is completed, the second fan shell 51 is mounted in the same way. The second outer cover 18 is first opened, and then the wire of the second fan assembly 5 is communicated to the second power supply assembly 9 in the second mounting chamber 1024 through the second conductive through hole 171. Then, the second outer cover 18 is closed. After the wire of the second fan assembly 5 is connected, the second fan shell 51 is assembled with the hanging main body 1 through the buckling between the connection buckles 315 and the connecting buckle slots 151, and finally a vertical blowing type neck fan that is conductive with the hanging main body 1 and easy to mount is obtained.

Through the above structure, after the assembly of the first fan shell 31 is completed, the second fan shell 51 is mounted in the same way. Firstly, the second outer cover 18 is opened. Then, the wire of the second fan assembly 5 is passed through the second conductive through hole 171 to be connected to the second power supply assembly 9 in the second mounting cavity 1024. Moreover, the second outer cover 18 is closed. After the wire of the second fan assembly 5 is connected, the second fan shell 51 is assembled with the hanging main body 1 through the buckling between the connection buckles 315 and the connecting buckle slots 151. Finally, a vertical blowing type neck fan that is conductive with the hanging main body 1 and easy to mount is obtained.

It can be understood that the inner sides of the first outer cover 16 and the second outer cover 18 are provided with at least one first snap-fit member 160, and the outer sides of the first main body portion 101 and the second main body portion 102 are provided with at least one second snap-fit member 1010. The first snap-fit member 160 and the second snap-fit member 1010 are snap-fitted to enable the assembly of the first outer cover 16 and the first main body portion 101 as well as the assembly of the second outer cover 18 and the second main body portion 102. Specifically, the number of the first snap-fit member 160 and the second snap-fit member 1010 can be multiple, such as 16a, 16b, 160c, 160d and 101a, 101b, 101c, 101d. Moreover, the structures of the multiple first snap-fit members 16a, 16b, 160c, 160d can be different, and the multiple second snap-fit members 101a, 101b, 101c, 101d are of the structures that match and cooperate with the multiple first snap-fit members.

It can be understood that both the first fan assembly 3 and the second fan assembly 5 are vertical blowing type fan assemblies with the air inlet and the air outlet on the same

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axis. In the first fan assembly **3**, the air inlet cover **31b**, the first fan blade **33**, the first motor **32**, the shell main body **31a** and the air outlet hood **31c** are all arranged in a straight line. In the second fan assembly **5**, the air inlet cover **31b**, the second fan blade **53**, the second motor **52**, the shell main body **31a** and the air outlet hood **31c** are all arranged in a straight line. In this way, the first fan assembly **3** and the second fan assembly **5** can generate relatively strong wind power and achieve a better air outlet effect.

Furthermore, according to Embodiment I and Embodiment II, it can be seen that the portable fan and the other portable fan can be respectively provided with corresponding and independent button switches **1015** and **1025**, so that the turning on and off of the portable fan and the other portable fan can be independently controlled by the corresponding button switches **1015** and **1025**.

Furthermore, referring to FIG. **25**, in a modified embodiment of Embodiment III, the air inlet holes **3110** of the metal barrier net **311** can also be in a regular hexagon. The plurality of air inlet holes **3110** are arranged in a honeycomb shape, and a surrounding wall **311a** is arranged between two adjacent air inlet holes **3110**. Moreover, Such a design can make the air inlet area of the metal barrier net **311** the largest, its weight the lightest, its wind resistance the smallest, and its air inlet volume the largest. Specifically, the width of the surrounding wall ranges from 0.05 mm to 2 mm.

As described above, one or more embodiments are provided in conjunction with the detailed description. The specific implementation of the present invention is not confirmed to be limited to that the description is similar to or similar to the method, the structure and the like of the present invention, or a plurality of technical deductions or substitutions are made on the premise of the conception of the present invention to be regarded as the protection of the present invention.

What is claimed is:

1. A portable fan, comprising:

- a first main body portion, wherein the first main body portion has an arc-shaped structure, and comprises a concave first air guide path positioned on a surface of one side of the first main body portion and extended along an extending direction of the arc-shaped structure;
- a first fan assembly, wherein the first fan assembly is detachably arranged at one end of the first main body portion and is configured to blow air towards the first air guide path, and the first air guide path is configured to guide air of the first fan assembly to one side of the first main body portion and one end of the first main body portion away from the first fan assembly;
- a second main body portion detachably connected to the first main body portion, wherein the second main body portion has an arc-shaped structure, and comprises a concave second air guide path positioned on a surface of one side of the second main body portion and extended along an extending direction of the arc-shaped structure; and
- a second fan assembly, wherein the second fan assembly is detachably arranged at one end of the second main body portion and is configured to blow air towards the second air guide path, and the second air guide path is configured to guide air of the second fan assembly to one side of the second main body portion and one end of the second main body portion away from the second fan assembly;

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wherein the first main body comprises a first air inlet arranged adjacent to one end of the first fan assembly, and a first air outlet arranged adjacent to one end of the first air guide path, the first air inlet and the first air outlet are arranged at a same central axis, and the first air guide path comprises a first air blowing opening aligned with the first air outlet; and

wherein the second main body comprises a second air inlet arranged adjacent to one end of the second fan assembly, and a second air outlet arranged adjacent to one end of the second air guide path, the second air inlet and the second air outlet are arranged at a same central axis, and the second air guide path comprises a second air blowing opening aligned with the second air outlet.

2. The portable fan according to claim 1, wherein the first fan assembly comprises a first fan shell, a first motor, and a first fan blade mounted on a rotating shaft of the first motor, the first motor and the first fan blade are arranged within the first fan shell; the first fan shell comprises a shell main body and an air outlet hood, the air outlet hood is arranged on one side of the shell main body; the air outlet hood comprises a plurality of air gathering rings and an annular wall structure connected to a periphery of the plurality of air gathering rings, a plurality of air outlet holes are formed among the plurality of air gathering rings, and the air gathering rings are configured to cooperate with an inner side surface of the annular wall structure to pressurize wind from the shell main body and then blow it out.

3. The portable fan according to claim 2, wherein the inner side surface of the annular wall structure is an inclined surface or an arc-shaped surface, so that an inner diameter of the annular wall structure gradually decreases along a direction from the shell main body to the air outlet hood.

4. The portable fan according to claim 2, wherein the air outlet hood further comprises a mounting structure connected to the plurality of air gathering rings, and the mounting structure comprises an annular mounting wall and a cover plate connected to the annular mounting wall.

5. The portable fan according to claim 4, wherein the mounting structure further comprises a mounting column connected to an inner side of the cover plate, and the mounting column is detachably connected to the shell main body; the cover plate is in a circular shape and is positioned in a middle of the plurality of air outlet holes, and a groove is arranged on an outer surface of the cover plate.

6. The portable fan according to claim 4, wherein the mounting structure is further provided with a first alignment structure, the shell main body has a second alignment structure, and the first alignment structure and the second alignment structure cooperate to enable an aligned assembly of the shell main body and the air outlet hood.

7. The portable fan according to claim 2, wherein the shell main body comprises an annularly arranged sidewall structure, an mounting portion positioned at one end inside the sidewall structure and an air guiding portion connected between the mounting portion and the sidewall structure, the air guiding portion comprises a plurality of air guiding blades arranged at intervals and is configured to guide wind from the first fan blade based on a preset direction, the first fan blade and the first motor are arranged within the sidewall structure and are positioned on one side of the mounting portion, and the first motor is mounted on the mounting portion.

8. The portable fan according to claim 4, wherein the mounting portion is provided with an opening, a conducting wire connected to the first motor extends through the open-

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ing, and then stretches into the first main body portion to be electrically connected with a first power supply assembly.

9. The portable fan according to claim 7, wherein the first fan shell further comprises a metal barrier net, the metal barrier net is arranged on one side of the sidewall structure away from the air outlet hood, the metal barrier net is provided with a plurality of air inlet holes, and an inner diameter of each the air inlet holes ranges from 0.5 mm to 4 mm.

10. The portable fan according to claim 9, wherein the air inlet hole is in a regular hexagon shape, a surrounding wall is arranged between two adjacent air inlet holes, and a width of the surrounding wall ranges from 0.05 mm to 2 mm.

11. The portable fan according to claim 9, wherein the first fan shell further comprises an air inlet cover, the metal barrier net is mounted on an inner side of the air inlet cover, the air inlet cover is arranged on one end of the shell main body away from the air outlet hood, the air inlet cover is in an annular shape, and the air inlet cover is sleeved on one end of the shell main body; an outer side surface of one end of the shell main body is provided with a stepped groove, the air inlet cover is sleeved and mounted on the stepped groove, and an outer side surface of the air inlet cover is flush with the outer side surface of the shell main body.

12. The portable fan according to claim 9, wherein the metal barrier net, the first fan blade, the first motor, the shell main body and the air outlet hood are all arranged in a straight line.

13. The portable fan according to claim 1, wherein the first main body portion is provided with a first mounting cavity, the portable fan further comprises a first power supply assembly and a first outer cover, the first power supply assembly is positioned within the first mounting cavity, the first mounting cavity is provided with an opening, and the first outer cover is detachably covered on the opening of the first mounting cavity.

14. The portable fan according to claim 13, wherein an inner side of the first outer cover is provided with at least one first snap-fit member, an outer side of the first main body portion is provided with at least one second snap-fit member, and the at least one first snap-fit member and the at least one second snap-fit member are snap-fitted to enable an assembly of the first outer cover and the first main body portion.

15. The portable fan according to claim 1, wherein the first fan assembly is detachably arranged on an inner side of the first main body portion, a plurality of first latching members are arranged on the inner side of the first main body portion, and the first fan assembly is provided with a plurality of second latching members detachably clamped with the first latching members.

16. The portable fan according to claim 1, wherein the first main body portion further comprises at least one air guiding vane, the at least one air guiding vane is arranged within the first air guide path adjacent to the first fan assembly, and is configured to guide the wind from the first fan assembly; the number of the at least one air guiding vane is two, the two air guiding vanes are respectively arranged on each of two sides of the first air guide path, and a distance between the two air guiding vanes gradually increases along an air outlet direction of the first air guide path.

17. A neck fan, wherein the neck fan comprises a first portable fan, a second portable fan and a connection member, the connection member is detachably connected to at least one of the first portable fan and the second portable fan assembly;

wherein the first portable fan comprises:

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a first main body portion, wherein the first main body portion has an arc-shaped structure, and comprises a concave first air guide path positioned on a surface of one side of the first main body portion and extended along an extending direction of the arc-shaped structure; and

a first fan assembly, wherein the first fan assembly is arranged at one end of the first main body portion and is configured to blow air towards the first air guide path, and the first air guide path is configured to guide air of the first fan assembly to one side of the first main body portion and one end of the first main body portion away from the first fan assembly;

wherein the first main body portion further comprises at least one air guiding vane, the at least one air guiding vane is arranged within the first air guide path adjacent to the first fan assembly, and is configured to guide the wind from the first fan assembly; the number of the at least one air guiding vane is two, the two air guiding vanes are respectively arranged on each of two sides of the first air guide path, and a distance between the two air guiding vanes gradually increases along an air outlet direction of the first air guide path.

18. The neck fan according to claim 17, wherein the connection member is detachably connected to the first portable fan and the second portable fan, the connection member comprises a flexible connection portion, the flexible connection portion is provided with at least one mounting hole, the at least one mounting hole is configured to be sleeved and fixed onto a first connection end and a second connection end, so that first main body portion and a second main body portion are connected through the flexible connection portion; the at least one mounting hole is an mounting through hole arranged along an extending direction, each of two ends of the mounting hole is respectively configured to sleeve and fix the first connection end and the second connection end; the first portable fan and the second portable fan are respectively provided with corresponding button switches, so that turning on and off the first portable fan and the second portable fan are independently controlled by the corresponding button switch.

19. A fan assembly, configured for being detachably connected to a main body portion of a neck fan, wherein the fan assembly comprises a first fan shell, a first motor and a first fan blade connected to the first motor, the first fan blade is arranged within the first fan shell, the first fan shell comprises a shell main body, an air outlet hood and a metal barrier net, the air outlet hood is detachably arranged on one side of the shell main body and is provided with an air outlet hole, the metal barrier net is arranged on the other side of the shell main body, the air outlet hood comprises a plurality of air gathering rings and an annular wall structure connected to a periphery of the plurality of air gathering rings, the metal barrier net comprises a plurality of air inlet holes that can reduce wind resistance and increase air intake volume, the plurality of air outlet holes are defined among the plurality of air gathering rings, and the air gathering rings are configured to cooperate with an inner side surface of the annular wall structure to pressurize wind from the shell main body and then blow it out,

wherein an inner side surface of the annular wall structure is an inclined surface or an arc-shaped surface, so that an inner diameter of the annular wall structure gradually decreases along a direction from the shell main body to the air outlet hood; and

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wherein each of the air outlet hood and the shell main body comprises a plurality of connection buckles for being detachably connected to the main body portion of the neck fan.

20. The portable fan according to claim 1, wherein the first main body portion comprises a first outer shell and a first inner shell detachably connected to first outer shell, the first air guide path is arranged in the first main body, the first air blowing opening is formed one end of the first inner shell; the second main body portion comprises a second outer shell and a second inner shell detachably connected to second outer shell, the second air guide path is arranged in the second main body, the second air blowing opening is formed one end of the second inner shell.

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